Multi-Layered Anti-Cheating Desktop Application for Secure Online Assessment using Environment Integrity and Behavioral Rule-Based Detection Algorithms

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Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Computer Science with Specialization in Software Engineering

at

FEU Institute of Technology

February 2026

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APPROVAL AND ACCEPTANCE SHEET

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ACKNOWLEDGMENT

(OPTIONAL)

The student may choose to include an acknowledgment page to thank those whom they wish to show gratitude to such as their parents, advisers etc. The content must be in English.

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Chapter 1

INTRODUCTION

1.1 Background of the Study

The COVID-19 pandemic has widely and radically changed the educational landscape all over the world, and institutions had no choice but to quickly switch from the traditional classroom setting to the online learning modality. One of the biggest challenges that came with this change was how to keep academic integrity during remote assessments, as teachers found it difficult to create the same conditions as the physical exam halls through digital platforms such as Canvas. Online examinations offered convenience and accessibility, but at the same time, they were highly vulnerable, students could secretly use unauthorized resources, have peer communication, or even apply advanced tech tools without being caught. Though Canvas and other Learning Management Systems offer basic proctoring features, they still inadequately handle the complex nature of dishonesty in offsite non-supervised remote settings. The absence of meticulous monitoring setups has raised serious doubts among academic institutions regarding the validity and reliability of online assessments, and thus the issue of academic credentials' credibility and the general integrity of distance education programs has surfaced again.

This program highlights and discusses these issues by devising a Multi Layered Anti Cheating Desktop Application which fuses real time integrity verification of the environment with the detection of behavioral anomalies using rule based algorithms to

form a comprehensive security structure for online assessments. The system, unlike the traditional single approach solutions, provides multiple layers of protection that keep track of both the testing environment and the student's behavior patterns in real time. The merging of environmental checks including unauthorized app detection or suspicious network activity detection with behavioral analysis that pinpoints irregular mouse movements, etc.

The swift transition to online learning has made assessments a lot more vulnerable to academic dishonesty techniques that are quite sophisticated and over the traditional proctoring tools that operate through browsers. Online exam systems usually function within the browser, thus they are unable to identify actions taking place at the system or application levels, examples of which include switching to a different window, using messaging or search applications, circumventing restrictions through virtual machines, or getting content out of the system through clipboard or screenshot tools. Moukhliss et. al. (2023) and Zheng et. al. (2024) state that the contemporary deception practices have made it imperative for the monitoring mechanisms to be adept in analyzing both user environments and their real-time device interactions. Otherwise, the exam platforms are left with no choice but to struggle in their attempts to authenticate the user's surroundings or catch the suspicious activities in real-time. It has been found by researchers that the usual monitoring techniques have become ineffective in the face of modern cheating practices and thus there is a need for setups that would not only direct student behavior but monitor their environment closely as well.

This study reacts to the aforementioned limitations by suggesting the creation of a Multi-Layered Anti-Cheating Desktop Application for Secure Online Assessment, which adopts two main approaches: Environment Integrity and Behavioral Rule-Based Detection. The Environment Integrity approach is carried out through the algorithms of Virtualization Artifact Check (VAC) and Hardware-Software Artifact Scan (HAS) that scrutinize the system files, hardware signatures, and recognized virtualization indications to figure out whether the pupil is using a virtual machine or emulator. The Behavioral Rule-Based Detection method is activated by the Real-Time Focus Monitoring (RTFM) algorithm for window-switch detection, Process Blacklisting Detection (PBD) for unauthorized use of applications, Clipboard and Screenshot Activity Detection (CSAD) for attempts to extract content, and the Behavioral Rule-Based Decision Engine (BRBDE), which ranks combined behaviors through a predefined ruleset to assess the aggressiveness of cheating.

To address the restrictions of browser-only applications, the system works as a committed desktop program. The objective of the system is to rebuild trust and moral values in online tests through a secure, reliable, and manageable examination atmosphere, which is supported by real-time surveillance, centralized session control, and the visualization of alerts on the instructor's side.

1.2 Theoretical Framework

The Defense in Depth (DiD) Strategy and Sykes and Matza's Neutralization Theory serve as the foundation for the development of the Multi-Layered Anti-Cheating Desktop Application. In a remote assessment setting, these theories offer the technical and psychological underpinnings required to guarantee academic integrity.

1.2.1. Neutralization Theory

The study's behavioral component is grounded in Gresham Sykes and David Matza's Neutralization Theory (1957). This theory explains that individuals who engage in dishonest behavior, such as academic cheating, do not necessarily reject societal norms; instead, they rationalize their actions through various justifications, commonly referred to as "neutralization techniques."

Application to the System: Based on the reviewed literature, the transition to the "new normal" in online education created conditions that made academic dishonesty easier to rationalize, with students often invoking explanations such as "the system is unfair" or "everyone is doing it." These forms of neutralization, particularly denial of responsibility and denial of injury, are addressed in the proposed system through objective, real-time monitoring. By recording observable actions such as window switching and clipboard usage, the system limits opportunities for post-exam justification and supports more accountable interpretation of student behavior during online assessments.

1.2.2. Multi-Layered Security and Environment Integrity

The concept of multi-layered monitoring, commonly referred to as the Defense in Depth (DiD) strategy in cybersecurity, serves as the foundation of this study's technical framework. This approach emphasizes that relying on a single security control is insufficient to preserve system integrity; instead, multiple complementary layers are implemented so that if one mechanism is bypassed, others remain capable of detecting and responding to potential threats.

Application to the System: This study addresses technical limitations identified by Zheng et al. (2024), who argue that browser-only monitoring mechanisms are inadequate for detecting modern cheating techniques. To overcome these limitations, the proposed system adopts a multi-layered, desktop-based architecture.:

- Layer 1: Environment Integrity: The system confirms the "physicality" of the workstation using the Hardware–Software Artifact Scan (HAS) and Virtualization Artifact Check (VAC). This stops unauthorized background activities from being concealed by Virtual Machines (VMs) or emulators.
- Layer 2: Continuous System Monitoring: Real-time detection demands monitoring outside of the active window, as noted by Essahraoui et al. (2022). The system locks down the operating system environment, ensuring the exam remains the sole focus of the hardware resources.

1.2.3. Behavioral Rule-Based Logic

Rule-Based Decision Theory serves as the foundation for the system used to identify and categorize potential academic dishonesty. Under this theory, a system applies logical If–Then rules to specific data inputs, enabling consistent classification of observed behavior patterns based on predefined conditions.

Application to the System: The proposed system employs an inference engine consistent with the Knowledge-Based Agent model to process data collected from student computer interactions during an examination session.

- Operationalization: The Behavioral Rule-Based Decision Engine (BRBDE) is used to do this. The system infers an intended violation by applying propositional logic to detected events, such as if Real-Time Focus Monitoring (RTFM) detects a window transition AND Clipboard & Screen Activity Detection (CSAD) detects a copy event.
- Classification: This enables the algorithm to discriminate between "Aggressive Violations" (planned cheating) and "Passive Violations" (accidental triggers) without the need for continuous human supervision, achieving the Detection Certainty suggested by Tiong and Lee (2024).

1.3 Conceptual Framework

This study utilizes the Input–Process–Output (IPO) model to structure the technical and operational flow of the proposed Multi-Layered Anti-Cheating Desktop Application. The IPO model serves as both a visual and logical representation of how raw system data, theoretical foundations, and technical requirements are transformed into a secure and verifiable online examination environment. Through this framework, the system is conceptualized as a knowledge-based security mechanism that evaluates multiple indicators of academic misconduct to support the study's objective of maintaining academic integrity in remote assessments.

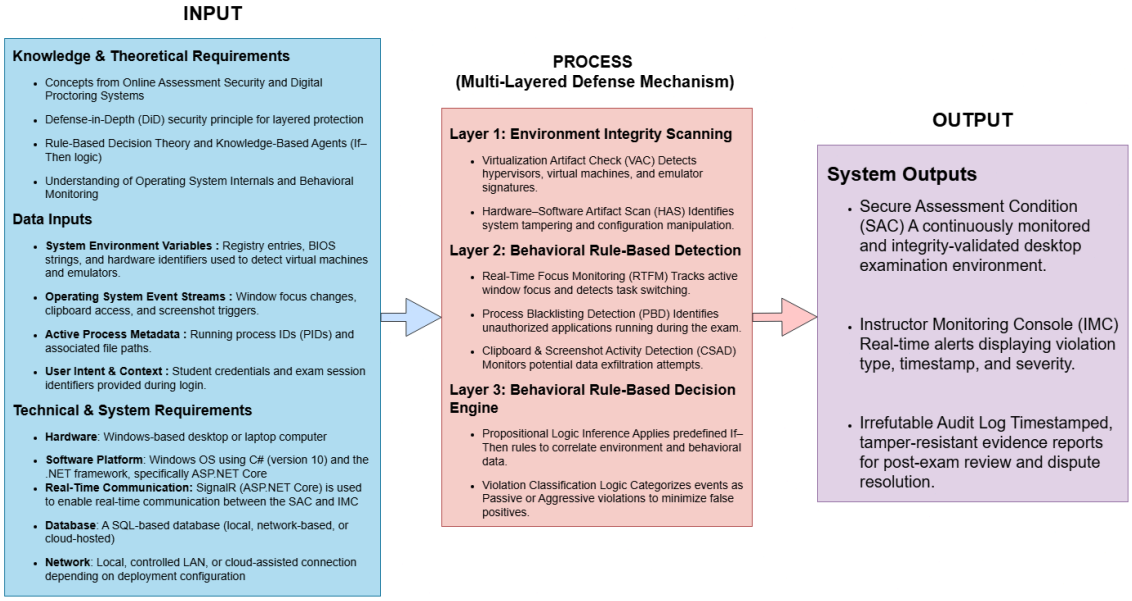

Figure 1. *Input-Process-Output*

1.3.1 Input

The Input component represents the theoretical foundations, technical data, and system requirements necessary for the proper operation of the anti-cheating system. These inputs

serve as the basis for all detection, evaluation, and decision-making processes within the system.

1.3.1.1 Knowledge and Theoretical Requirements

- The system is grounded in established concepts from **online assessment security and digital proctoring systems**, which emphasize the need for continuous monitoring beyond browser-level restrictions. These concepts inform the design of monitoring mechanisms that operate at the operating system level rather than relying solely on surface-level controls.
- The framework also incorporates the **Defense-in-Depth (DiD)** security principle, which advocates the use of multiple, overlapping security layers to prevent system compromise. Instead of relying on a single detection mechanism, the system applies layered environment and behavioral checks to ensure that weaknesses in one layer do not result in total system failure.
- In addition, the system is based on **Rule-Based Decision Theory and Knowledge-Based Agent principles**, where predefined If–Then logical rules are applied to system data to derive conclusions. This enables the system to reason about complex behaviors and distinguish between normal and suspicious activities in an automated and consistent manner.
- A working understanding of operating system internals and behavioral monitoring further supports the system's ability to interpret low-level events such as process execution, window focus changes, and clipboard interactions, which are commonly exploited during online examinations.

1.3.1.2 Data Inputs

To operationalize these theoretical foundations, the system collects several categories of technical data.

- **System Environment Variables**, including registry entries, BIOS strings, and hardware identifiers, are gathered to verify the authenticity of the examination environment. These variables are essential for detecting the presence of virtual machines, emulators, or simulated environments that may be used to bypass exam restrictions.
- **Operating System Event Streams**, such as window focus changes, clipboard access events, and screenshot triggers, are continuously monitored to capture real-time user behavior. These events provide behavioral indicators that may suggest task switching, information copying, or unauthorized data capture during the examination.
- **Active Process Metadata**, consisting of running process identifiers (PIDs) and associated file paths, is collected to identify unauthorized or blacklisted applications executing in the background. This allows the system to detect the use of secondary browsers, communication tools, or automation software during the exam session.
- **User Intent and Context data**, including student credentials and exam session identifiers, are used to bind all detected events to a specific user and examination instance. This contextualization ensures accurate attribution of activities and supports reliable post-exam review.

1.3.1.3 Technical and System Requirements

- The system is designed to operate on Windows-based desktop or laptop computers, reflecting the target environment for secure online examinations. The Secure Assessment Client (SAC) and Instructor Monitoring Console (IMC) are implemented using the C# programming language (C# version 10) and the .NET framework, specifically ASP.NET Core, which enables robust application development and real-time communication. This platform allows controlled interaction with operating system–level services and event streams through user-space APIs, without requiring kernel-level access or system modification.
- To support real-time communication between the student-side SAC and the teacher-side IMC, the system utilizes SignalR, a library within ASP.NET Core that enables bi-directional, real-time data exchange. SignalR is used to transmit monitoring events, violation alerts, timestamps, and severity levels, ensuring that instructors receive timely updates during examination sessions.
- A local, network-based, or cloud-hosted SQL database is used to store system logs, detected events, and audit records in a structured and persistent manner. During prototype implementation, data storage and communication may operate within a local or controlled LAN environment to ensure reliability and reduce external dependencies. For extended deployment scenarios, the architecture supports cloud-based servers to enable centralized monitoring, scalability, and multi-site access, provided that secure communication protocols and data protection measures are applied.

- Network communication is designed to be configurable, allowing the system to function in local, LAN-based, or cloud-assisted modes, depending on deployment requirements. This flexible architecture supports both controlled testing environments and future large-scale institutional deployment while maintaining security, reliability, and data integrity during active examinations.

1.3.2 Process

The Process component represents the core detection and decision-making workflow of the system. It consists of a multi-layered defense mechanism that sequentially analyzes input data using specialized algorithms and logical rules.

1.3.2.1 Layer 1: Environment Integrity Scanning

- The first layer focuses on verifying the legitimacy of the examination environment.
- The **Virtualization Artifact Check (VAC)** examines system indicators to detect the presence of hypervisors, virtual machines, or emulator signatures. This prevents examinees from running the exam within simulated environments designed to hide unauthorized activities.
- The **Hardware–Software Artifact Scan (HAS)** analyzes system configurations and hardware–software interactions to identify signs of tampering or manipulation intended to bypass monitoring hooks. Together, VAC and HAS ensure that the exam is conducted on a genuine and uncompromised computing platform.

1.3.2.2 Layer 2: Behavioral Rule-Based Detection

The second layer monitors examinee behavior during the examination session.

- **Real-Time Focus Monitoring (RTFM)** tracks the active application window to detect task switching away from the examination interface, which may indicate access to unauthorized resources.
- **Process Blacklisting Detection (PBD)** continuously checks running processes against a predefined blacklist of unauthorized applications, enabling detection of prohibited tools such as messaging platforms or secondary browsers.
- **Clipboard and Screenshot Activity Detection (CSAD)** monitors clipboard usage and screenshot triggers to identify potential attempts at copying exam content or capturing sensitive information for later use.

1.3.2.3 Layer 3: Behavioral Rule-Based Decision Engine

The final layer functions as a decision-making engine that evaluates data collected from the previous layers.

- Through **Propositional Logic Inference**, predefined If–Then rules are applied to correlate environmental and behavioral indicators. This logical reasoning allows the system to determine whether detected events represent benign system behavior or potential academic misconduct.
- The **Violation Classification Logic** categorizes detected events as Passive Violations (system- or environment-driven anomalies) or Aggressive Violations (intentional user actions). This classification minimizes false positives and ensures fair interpretation of detected activities.

1.3.3 Output

The Output component represents the system's responses and deliverables that support exam integrity and accountability.

1.3.3.1 System Outputs

- The primary output is the **Secure Assessment Condition (SAC)**, defined as a continuously monitored and integrity-validated desktop examination environment. Rather than forcibly restricting user actions, the system maintains exam integrity by detecting, evaluating, and recording relevant system and behavioral events.
- The **Instructor Monitoring Console (IMC)** provides real-time alerts to instructors, displaying the type of detected violation, its timestamp, and severity level. This enables timely awareness and informed decision-making during or after the examination.
- Finally, the system generates an **Irrefutable Audit Log**, consisting of timestamped and tamper-resistant evidence reports. These logs provide objective documentation for post-exam review and dispute resolution, strengthening transparency and trust in the online assessment process.

1.4 Statement of the Problem

This study seeks to answer the following research questions:

1. How can online assessment systems identify and verify the legitimacy of computing environments during remote examinations without requiring kernel-level access?
2. How can unauthorized or suspicious user behaviors be detected and recognized during online assessments, and how can such behaviors be distinguished from unintentional actions?
3. How can instructors be supported in monitoring online examinations through real-time visibility and session management mechanisms?
4. How can secure and reliable communication be ensured between student assessment platforms and instructor monitoring tools throughout an online examination session?
5. How can user trust and perceived system usability be evaluated when comparing traditional online assessments with online assessments enhanced with additional integrity monitoring mechanisms?
6. How can the quality characteristics of an online assessment security system be assessed in accordance with recognized software quality standards within an educational context?

1.5 Objectives of the Study

General Objectives

The primary objective of this thesis is to design, develop, and implement a secure, multi-layered anti-cheating system that functions as a dedicated desktop application to address the security limitations of browser-based online proctoring.

Specific objectives:

1. To develop and implement an installable desktop application, the Secure Assessment Client (SAC), capable of executing multi-layered, application-level monitoring through the integration of Virtualization Artifact Check (VAC) and Hardware–Software Artifact Scan (HAS) algorithms to detect virtual machines, emulators, unauthorized device environments and activities.

2. To evaluate the capability of the Secure Assessment Client (SAC) in detecting unauthorized activities by analyzing the desktop status through the Behavioral Rule-Based Decision Engine (BRBDE); this engine integrates data from RTFM, PBD, and CSAD, utilizing propositional logic to classify user desktop activities into passive or aggressive violations.

3. To provide educators with a dedicated Instructor Monitoring Console (IMC) capable of creating secure sessions and visualizing real-time alerts transmitted by connected student clients.

4. To design and deploy a Proctoring Session Server (PSS) that facilitates secure communication, manages session synchronization, and relays monitoring data between the SAC and IMC.

5. To evaluate and compare user trust levels and perceived system usability between a traditional online assessment and a traditional online assessment integrated with the SAC and IMC, using the System Usability Scale (SUS) and a Likert-scale questionnaire to assess system acceptance.

6. To assess the system's quality characteristics and perceived usability based on ISO/IEC 25010 standards and the System Usability Scale (SUS) in an educational assessment context.

1.6 Scope and Delimitations

The scope of this study focuses on the design, development, and evaluation of a Client-Server Multi-Layered Anti-Cheating System implemented through standalone Windows desktop applications for Windows 10 and above. The system architecture includes a Secure Assessment Client (SAC) for student monitoring, an Instructor Monitoring Console (IMC) for real-time session management, and a Proctoring Session Server (PSS) for data relay. Functional implementation is limited to application-level

monitoring using Real-Time Focus Monitoring (RTFM), Process Behavior Detection (PBD), Command/Shortcut Activity Detection (CSAD), and a Behavioral Rule-Based Decision Engine (BRBDE), alongside environment integrity tools such as the Virtualization Artifact Checker (VAC) and Hardware Authenticity Scanner (HAS).

However, the study is delimited to the Windows environment and does not support mobile, tablet, or non-Windows devices. The system is strictly standalone and does not feature integration with Learning Management Systems (LMS) such as Canvas or Moodle. Furthermore, the detection boundary is restricted to the device running the software; it cannot identify cheating via external secondary devices, offline collaboration, or hardware manipulation. Finally, the system does not utilize biometric or visual proctoring features like face recognition, eye-tracking, or video recording, and it operates without kernel-level drivers, which may limit its ability to block highly privileged system-level operations.

1.7 Significance of the Study

The outcomes of this study are especially significant for the following stakeholders:

Educational Institutions. This study provides educational institutions with a structured approach for conducting online examinations by addressing limitations commonly associated with browser-based proctoring systems. The proposed multi-layered anti-cheating desktop application supports the conduct of more controlled and reliable remote assessments, contributing to greater confidence in online evaluation processes and academic credentials.

Students. The study promotes fairness in online assessments by discouraging dishonest practices and encouraging compliance with examination rules. By establishing a transparent and monitored testing environment, students are guided toward responsible assessment behavior, allowing examination results to more closely reflect individual performance.

Educator and Instructors. Educators and instructors are supported through real-time monitoring and centralized session control that provide visibility into assessment activities. The system assists instructors in identifying and reviewing suspicious behaviors, helping reduce uncertainty in assessment outcomes while supporting fair and consistent evaluation practices.

Research and Developers. This study contributes to the fields of computer science and educational technology by presenting a system design that integrates environment integrity verification with behavioral rule-based detection. The proposed framework may serve as a reference for future research and development of secure online assessment systems.

Online Assessment System Designers. System designers may gain practical insights into implementing multi-layered security approaches that extend beyond browser-level monitoring. The proposed desktop-based architecture demonstrates how combining

environment checks with behavioral analysis can address contemporary challenges in online assessment security.

1.8 Definition of Terms

The following terms are defined according to their specific functions and conceptual roles within the scope of this study:

Application-Level Monitoring - Refers to the surveillance of software activities that occur within the user's operating system environment without requiring deep, kernel-level access to the computer's core.

Behavioral Rule Based Decision Engine (BRBDE) -The core logic component of the system that applies "if-Then" rules to distinguish between accidental actions and intentional cheating.

Clipboard and Screenshot Activity Detection (CSAD) - A monitoring feature designed to detect when a student attempts to copy exam content or take a screen capture of the assessment.

Defense in Depth (DiD) - A cybersecurity strategy used in this study that employs multiple layers of security (Environment and Behavior) to ensure that if one layer is bypassed, another will detect the violation.

Environment Integrity - The state of a student's digital workspace being "clean" and free from unauthorized tools like virtual machines of third-party cheating software.

Hardware-Software Artifact Scan (HAS) - A technical check that analyzes hardware signatures and system files to ensure the student is using a genuine physical computer.

Instruction Monitoring Console (IMC) - The dedicated desktop application used by teachers to manage exam sessions and receive-real-time alerts of suspicious student behavior.

Multi-Layered Anti-Cheating System - A security architecture that combines different detection methods to provide a more robust defense than a single-method tool.

Neutralization Theory - A psychological framework explaining how students use justifications role "denial of responsibility" to silence their guilt when committing academic dishonesty.

Online learning Modality - The educational setup where teaching and assessments are conducted remotely via the internet rather than in a physical classroom.

Proctoring Session Server (PSS) - The backend infrastructure that manages the communication, data relay, and synchronization between the student's client and the instructor's console.

Process Blacklisting Detection (PBD) - A security feature that monitors for the execution of forbidden applications, such as web browsers, messaging apps, or search tools, during an active exam.

Real-Time Focus Monitoring (RTFM) - A mechanism that tracks whether the exam window remains the "active" window on the student's screen, detecting when a student switches to another program.

Secure Assessment Client (SAC) - The desktop application installed on the student's device that executes all monitoring and integrity checking algorithms.

Virtualization Artifact Check (VAC) - A specific algorithm used to detect if the exam is being run inside a Virtual Machine or emulator to bypass security restriction

Chapter 2

REVIEW OF RELATED LITERATURES

2.1 Foreign Literatures

2.1.1 Online Assessment Security in Digital Learning

The integrity of academic assessments has been seriously threatened by the quick transition to online learning. Online tests give students more access to digital tools, outside materials, and third-party apps that could be used for cheating than traditional classroom exams.. emphasized the lack of physical supervision in online settings has made it more challenging to maintain academic integrity. As a result, more sophisticated technical solutions are needed to protect assessment procedures and guarantee examinee fairness Shivanagi et al (2023).

2.1.2 Limitations of Browser-Based Proctoring Systems

The majority of online test systems use browser-based proctoring technologies to limit what students can do while taking tests. However, Bertiz and Hebebci noted that these systems can only keep an eye on activities carried out within the browser window only. Operating system-level activities, such background programs, supplementary displays, or external software, frequently go unnoticed. Because of this, browser-based solutions are becoming less effective at stopping complex types of academic dishonesty (Bertiz and Hebebci 2021).

2.2.2.1 Need for Desktop-Based Assessment Solutions

Desktop-based evaluation solutions provide more control over the whole computer environment, enabling more thorough examination of user behavior and system operations. Desktop solutions can monitor unapproved applications, identify unusual activity outside of the browser, and impose more stringent exam conditions because they function at the system level. Compared to browser-only proctoring solutions, these features make desktop-based platforms more successful in combating contemporary cheating strategies (Bertiz & Hebebci, 2021).

2.1.3 Environment Integrity and Virtualization Detection

Ensuring that the examinee's digital workspace is safe and devoid of unapproved technologies like virtual machines, emulators, or covert system processes is the main goal of environment integrity. Students may alter their system configurations to get around exam limits or hide cheating if there are no environment-level controls in place. According to studies of Vaezi , a crucial element of safe online evaluation systems is preserving control over the digital environment.

2.1.3.1 System-Level Monitoring for Secure Environments

During online tests, system-level monitoring allows for ongoing examination of software and hardware artifacts to spot questionable settings. Exam systems can stop students from taking advantage of system flaws by identifying abnormalities like virtualization indicators or unapproved background services. By addressing risks that browser-based

monitoring techniques usually miss, this strategy improves the security of online exams (Vaezi et al. 2024).

2.1.4 Behavioral Rule-Based Detection in Online Exams

Behavioral rule-based detection looks for possible academic misconduct through analysis of digital behavior patterns rather than physical observation. This strategy is based on the idea of Knowledge-Based Agents, which are automated systems that use logic to analyze data. Russell and Norvig (2020) claim that knowledge-based agents draw inferences about complex actions using an inference engine that assesses data using propositional if-then rules. This reasoning approach allows systems to evaluate behavioral signs like clipboard usage, irregular interaction timing, and frequent window switching in online exams. According to Kumar and Rohil (2024), keeping an eye on these digital behaviors makes it possible to automatically identify suspicious activity, which lessens the need for human oversight and makes the method appropriate for extensive online evaluations.

2.1.4.1 Activity Logging and Behavioral Analysis

Exam systems can record screenshots, user interactions, and system-level events during the exam thanks to continuous activity logging. By applying pre-established behavioral rules to this logged data, suspicious activity patterns can be found and a trustworthy audit trail for post-exam analysis is produced. According to the study by Kumar and Rohil, this approach supports reasonable and fact-based decision-making in cases of possible academic misconduct while improving transparency and accountability.

2.1.5 Multi-Layered Architecture for Exam Security

Researchers point out that integrating multiple levels of detection increases robustness against cheating, as relying on a single security technique is insufficient. This aligns with the design concept of desktop-based multi-layered anti-cheating systems, which combine behavioral monitoring and environment integrity to maximize the reliability of online exams Sakhipov (2025).

2.1.5.1 Multi-Layered Architecture for Exam Security

The Defense in Depth approach, which entails implementing several overlapping security measures to safeguard system integrity, is emphasized in modern security architectures. The NIST Special Publication 800-53 suggests layered protections to guarantee that a system cannot be compromised by a single control failure, This idea is consistent with the conclusions of Sakhipov , who contend that maintaining integrity in online tests requires more than just one security method. A desktop-based anti-cheating system adheres to the Defense in Depth strategy by integrating behavioral monitoring techniques and layered environment integrity checks. This ensures that even if one detection layer is circumvented, other controls are still in place to identify and neutralize any threats.

2.1.6 Development of an Intelligent Anti-Cheating System for Secure Online Examinations

A foreign study conducted by Tuloli, Latief, and Rohandi (2021) focused on addressing the increasing issue of cheating in remote online examinations brought about by rapid technological advancements. The researchers emphasized that traditional monitoring

approaches are no longer sufficient to ensure academic integrity in fully online assessment environments.

To address this concern, the study developed an intelligent anti-cheating system that integrates advanced monitoring techniques such as head posture recognition and eye-tracking analysis to observe student behavior during examinations. By continuously analyzing behavioral patterns, the system is capable of identifying suspicious actions that may indicate potential cheating. In addition to behavioral monitoring, the system also incorporates several security mechanisms including screenshot prevention, clipboard restriction, and remote access blocking to minimize opportunities for unauthorized assistance.

The findings of the study revealed that the proposed system was effective in detecting and preventing various forms of cheating while improving the overall efficiency of examination supervision. The results highlight the importance of implementing advanced and multi-layered technological solutions to promote fairness, credibility, and security in modern online assessment systems.

2.1.7 Design and Implementation of an Anti-Cheating System for Online Examinations

A foreign study conducted by Zheng, Lin, and Chen (2024) addressed the growing concern of academic dishonesty in remote online examinations brought about by continuous technological advancements. The researchers emphasized that traditional monitoring methods are becoming less effective in fully online assessment environments, highlighting the need for more advanced and reliable examination supervision systems.

The proposed system integrates intelligent monitoring technologies such as head posture recognition and eye-tracking analysis to observe student behavior during examinations. Through continuous behavioral analysis, the system can identify suspicious actions that may indicate possible cheating. In addition to behavioral monitoring, the system incorporates security features including screenshot prevention, clipboard restriction, and remote access blocking to minimize the risk of external assistance during examinations.

The findings revealed that the system was effective in detecting and preventing various forms of cheating while improving the overall efficiency and fairness of exam monitoring. The study further emphasized the importance of adopting advanced and multi-layered technological approaches to strengthen the integrity, security, and credibility of online examination systems.

2.2 Local Literatures

2.2.1 Outsourcing and Lack of Discipline

When the pandemic brought significant changes in education in students' lives, the world has shifted to a modern learning process toward online education, the issue of academic dishonesty, it did not conclusively claim that cheating is more prevalent online than in face-to-face classes, Using the data from the 650 members of social media groups that promote academic services where students can can outsource their academic task such as quizzes,projects , exams and assignments, the finding revealed a lack of discipline among students,particularly those in 2020 - 2021 academic year. These results show the growing challenge of maintaining academic integrity in online assessments, (Aguilar, 2022).

Overall, the study provides strong evidence that academic dishonesty remains a critical issue in online education, particularly at higher educational levels. The study strengthens the justification for implementing secure, intelligent, and behavior-aware assessment systems to preserve academic integrity in online examinations, which aligns directly with the objectives of the proposed multi-layered anti-cheating desktop application for Secure Online Assessment using Environment Integrity and Behavioral Rule-Based Detection.

2.2.2 Limitations of Policy and Traditional Monitoring

The researchers carried out a descriptive analytical study at a higher education institution in Manila to investigate students' perceptions of academic integrity and their tendencies toward dishonest academic behavior in an online learning environment. The research utilized McCabe's Academic Integrity Survey Report and collected data from 360 first and second year college students. The results indicated that while institutional policies on academic integrity were seen as well-defined and effectively communicated there was only a slight positive correlation between students' awareness of these policies and their propensity to engage in specific types of academic dishonesty Lichauco et. al., 2023).

Furthermore, the study highlights that plagiarism-detection software can reduce specific types of academic misconducts that still exist; it does not fully address other cheating behaviors that occur on online assessments, such as impersonation, collaboration with other students during online assessments or the use of external resources.

In relation to the present study, the findings support the development of a multi-layered anti cheating desktop application that combines environment integrity monitoring and behavioral rule based-detection. real-time behavioral and system level indicators of cheating , thereby strengthening the security and credibility of online assessments.

2.2.3 Behavioral Factors and the Ease of Cheating

The researchers studied the shift from face-to-face to online education during the COVID-19 pandemic and found that lenient academic policies and limited supervision significantly increased the academic dishonesty during online assessments. Using large scale surveys, that focus group discussion, and interviews, the study revealed that a majority of the students admitted to cheating during online assessments and perceive it as easier and less likely to be detected with behavioral and assessment related factors identified as the primary causes. The highlight of the study is the vulnerability of online assessment formats, particular objective type exams.Amorado & Vilchez, (2023).

This extensive study on Philippine students found that behavioral factors were prevalent causes of cheating and that a majority of students found it easier to cheat online, directly supporting the focus of "multi-layered anti-cheating desktop application that integrates environment integrity monitoring and behavioral rule-based detection" to address both behavioral tendencies and system-level weaknesses in secure online assessments.

2.2.4 The Need for Innovative Technical Solutions

Explored academic integrity through a mixed-method analysis involving surveys and structured interviews. The results indicated that while lecturers often observed cheating, they frequently hesitated to impose severe punishment. Qualitative data suggested that

students cheat due to factors such as procrastination, time constraints, and lack of comprehension, despite being aware of ethical standards. Significantly, the students themselves emphasized the need for institutional innovation to curb misconduct Jose (2022).

This study provides evidence that cheating in online examinations is a significant issue in Philippine colleges. The tendency of students to deny responsibility for their actions highlights the need for systems that minimize or eliminate opportunities for cheating. In response to this issue, the proposed Multi-Layered Anti-Cheating Application, which detects unauthorized tools and utilizes behavioral rule-based detection to monitor real-time actions, aims to remove the opportunities that enable students to deny responsibility for academic misconduct.

2.2.5 Impact of the New Normal on Academic Integrity and Student Behavior

The study was conducted at the Polytechnic College of Botolan, where the researchers investigated the impact of the "new normal" education system on students' academic integrity. The findings revealed that plagiarism and cheating during online examinations were the most common forms of academic misconduct. Using a descriptive-correlational research design, the study involved 150 students and 30 instructors. Anchored on Sykes and Matza's Neutralization Theory, the researchers found that students frequently employed denial of responsibility to justify their dishonest behaviors. Overall, the results indicated that remote learning has adversely affected student integrity. (Sonajo, 2022).

This study provides evidence that cheating in online examinations is a major issue in Philippine colleges. The tendency of students to deny responsibility for their actions

highlights the need for a system that minimizes or removes opportunities for academic dishonesty. In response, the proposed Multi-Layered Anti-Cheating Application, which integrates unauthorized tool detection and behavioral rule-based monitoring, aims to observe students' real-time actions during assessments. By limiting opportunities to cheat, the system addresses the core issue that allows students to rationalize and deny responsibility for their academic misconduct.

REVIEW OF RELATED STUDIES

2.3 Foreign Studies

2.3.1 Cheating in Online Examination Systems

The rapid growth of online education has also increased concerns regarding cheating during online examinations. According to , the shift from traditional classroom exams to online exams has created many challenges in maintaining academic integrity. The study reviewed different forms of cheating, common cheating methods, and detection techniques used in online examination systems. It discussed the use of technologies such as machine learning, computer vision, and remote proctoring to address cheating issues. The study emphasized that increased access to online resources has made cheating easier for students. It also highlighted that many existing systems still struggle to detect advanced cheating techniques. The findings show that there is a strong need for improved and more reliable anti-cheating solutions in online assessments (Dhamu, 2024).

2.3.2 Real-Time Monitoring and Rule-Based Detection Systems

Several foreign studies proposed systems that use real-time monitoring and rule-based detection to reduce cheating during online exams. developed an E-Exam Cheating Control System that combines webcam monitoring, screen activity tracking, keyboard and mouse behavior logging, and network inspection. The system uses both rule-based logic and artificial intelligence to detect suspicious behaviors such as switching exam windows, presence of multiple faces, and unauthorized device usage. The results showed that the system achieved over 90% detection accuracy in controlled environments. Derrick et al. (2025).

2.3.3 Automated and AI-Based Online Proctoring Systems

Automated proctoring systems using artificial intelligence have been widely studied. proposed AutoOEP, a multi-modal proctoring framework that uses computer vision and machine learning. The system analyzes facial movements, gaze direction, head pose, and hand activity using deep learning models such as CNNs and LSTMs. The results showed high accuracy in detecting suspicious activities. While these systems are effective, they depend on video recording and complex models, which may not be suitable for all examination environments (Naveen et al., 2025).

2.3.4 Browser-Based Cheating Detection Approaches

Browser-based cheating detection has also been explored in foreign studies. developed a browser plugin that tracks eye gaze movements using a webcam during online exams. The system analyzes eye gaze patterns and uses a One-Class Support Vector Machine to detect abnormal behavior that may indicate cheating. While the study showed that eye tracking can help detect suspicious behavior, browser-based solutions are limited to activities inside the browser. Actions such as running external programs or system-level manipulation may not be detected, which limits the effectiveness of browser-only solutions (Dilini et al., 2021).

2.3.5 The Evolution of Remote Proctoring System

A foreign study introduced a multimodal authentication system designed to strengthen academic integrity during online exams by combining identity verification, continuous facial recognition, audio analysis, and object detection. The system authenticates students before the test and monitors their presence and behavior in real time, detecting suspicious sounds and prohibited objects while automatically recording violations with timestamps

and screenshots. Involving 30 first-year Computer Department students from Mansoura University, the study found a significant decrease in exam scores after the system's implementation ($p < 0.01$), with a large effect size indicating reduced cheating and more consistent, honest performance. These findings support the effectiveness of integrating layered biometric and behavioral monitoring an approach that directly aligns with the goals of our own multilayered anti-cheating system Esmat et al., (2025).

2.4 Local Studies

2.4.1 Technology-Aided Cheating in Online Distance Learning

The internet and distance learning have made it really tough for schools to make sure students are honest with their work. This is a problem, in classes that are only online because teachers cannot watch students directly. Ravasco said that it is easier for students to cheat in distance learning classes because technology helps them do it. Someone did a study to see what students think about cheating, how they cheat and how to stop it at the University of the Philippines Open University. They wanted to know how often students cheat, what methods they use to cheat and what the University of the Philippines Open University can do to prevent cheating. Ravasco did a study. Found out that a lot of students think there is more cheating in online classes than in regular classes where you see the teacher in person. Three fourths of the people who answered the study feel that it is easier to cheat in online classes. This is because they can use a lot of resources and they do not have to talk to the teacher. The study also found out that students use technology to cheat in ways, such as pretending to be someone else using technology when they are not supposed to and copying work from the internet. study

shows that students use these methods like pretending to be someone which happened about seventy three percent of the time and using technology when they are not supposed to which happened about forty eight percent of the time and they also copy work from the internet. These results emphasize the urgent need for multiple security measures that target both the integrity of the environment and behaviors to tackle technology facilitated cheating in online assessments.Ravasco's (2021).

2.4.2 AI-Surveillance System for Examination Cheating Prevention

The old ways of watching students during tests are not working well. So people are looking into using intelligence to make sure students are honest during assessments. According to Almaden et al(2025). that the usual ways of catching cheaters depend on people watching and supervising which is not very good. These old ways are time consuming, hard to keep doing all the time and rely on peoples opinions, which can be different from one person, to another. This means that cheating is not always caught. The study made a system that uses intelligence to watch students and it is based on something called the YOLOv8 object detection framework. Artificial intelligence is used to help with integrity during assessments and the YOLOv8 object detection framework is a key part of this artificial intelligence system. This system looks at how people behave and finds things that're not normal to figure out if someone is doing something bad during exams. The people studying this focused on four things that people do: looking to the left, looking to the right, leaning to the left and leaning to the right. These things are often connected to cheating. The system was tested with a lot of pictures. 3,566 Images with 10,387 notes on them. It did a good job getting it right 82.4% of the time. The system was good at finding some of these behaviors like looking left and looking right. It was right

about these things between 75.2% and 88.5% of the time. The behaviors, like leaning left and leaning right were also found by the system with accuracy. These results demonstrate the effectiveness of rule-based detection methods for identifying specific cheating-related movements in real time. They also highlight the need for layered security approaches that combine behavior analysis with monitoring of the exam environment in online assessments Almaden et al(2025).

2.4.5 Academic Integrity in the New Normal Education

According to a study done at Polytechnic College of Botolan in the Philippines. The study looked at what 30 instructors and 150 students thought about how this new way of learning affects integrity.

The people who did the study used a theory called Neutralization Theory by Sykes and Matza from 1957 to analyze the data. They also used some methods like the Likert Scale and things, like Weighted Mean and Frequency and Percentage Distribution and something called Pearson. The results were clear. Students, in this new way of learning are most likely to cheat by copying someone else's work, which is called plagiarism. After that they cheat on tests. When students get caught they usually say it is not their fault. They say they had much work to do or the assignments were too hard. What the study found out is that when students think it is okay to cheat they will make excuses for it. This means that learning in this way is actually making students less honest when it comes to school work. The new normal education is affecting students academic integrity in a way. These findings emphasize the critical need for Multi-Layered Anti-Cheating Desktop Applications that integrate technological solutions for environment monitoring

and behavioral detection to address the complex factors driving academic dishonesty in online assessment contexts, particularly when students rationalize misconduct through external justifications.Sonajo, M. R. D. (2022).

2.5 Synthesis

The reviewed foreign and local literature consistently show that academic dishonesty in online assessments remains a serious concern. Students often perceive online examinations as easier to manipulate due to limited supervision, increased access to digital resources, and reduced monitoring (Shivanagi et al., 2023; Aguilar, 2022; Amorado & Vilchez, 2023). Behavioral factors such as rationalization and denial of responsibility further contribute to dishonest conduct in remote learning environments (Sonajo, 2022). These findings indicate that both technological vulnerabilities and behavioral tendencies play significant roles in online cheating.

Most existing anti-cheating systems rely on browser-based monitoring, webcam surveillance, or artificial intelligence-driven detection (Dilini et al., 2021; Naveen et al., 2025; Derrick et al., 2025). Although these approaches can detect certain suspicious behaviors, they are generally limited to activities within the browser or camera frame (Bertiz & Hebebci, 2021). Studies emphasize that such systems often fail to detect system-level manipulation, including the use of unauthorized applications, background processes, or virtualization tools (Vaezi et al., 2024). This limitation reveals a gap in ensuring comprehensive security during online examinations.

To address this gap, the proposed Multi-Layered Anti-Cheating Desktop Application integrates environment integrity mechanisms with behavioral rule-based detection. By focusing on desktop-level monitoring rather than relying solely on video or browser-based tools, the system offers a more comprehensive yet less intrusive approach to identifying suspicious activities. This layered strategy aligns with recommendations for stronger security architectures in online assessments (Sakhipov, 2025) and contributes to improving the reliability and fairness of remote examination environments.

Chapter 3

METHODOLOGY

3.1 Research Type

This study adopts an applied research type using a descriptive–comparative research design to address the growing challenge of maintaining academic integrity, usability, and user trust in online assessment environments. The research problem centers on evaluating how users perceive system usability and trust when comparing a traditional browser-based online assessment with an online assessment integrated with a multi-layered anti-cheating desktop application. Applied research is appropriate because the study focuses on developing and assessing a practical technological solution intended for real-world educational use rather than producing purely theoretical outcomes. The descriptive–comparative design enables systematic description and comparison of user perceptions across two assessment conditions without manipulating variables or employing inferential experimentation, consistent with established descriptive research frameworks (Client Challenge, n.d.-b; Romeyn, n.d.). This approach guides system development by emphasizing user-centered evaluation after actual system use, ensuring that design decisions align with usability and acceptance objectives. Validation and evaluation are supported through standardized survey instruments and descriptive statistical analysis, allowing meaningful comparison of outcomes while maintaining methodological rigor and practical relevance.

3.2 System Design

3.2.1 Activity Diagram

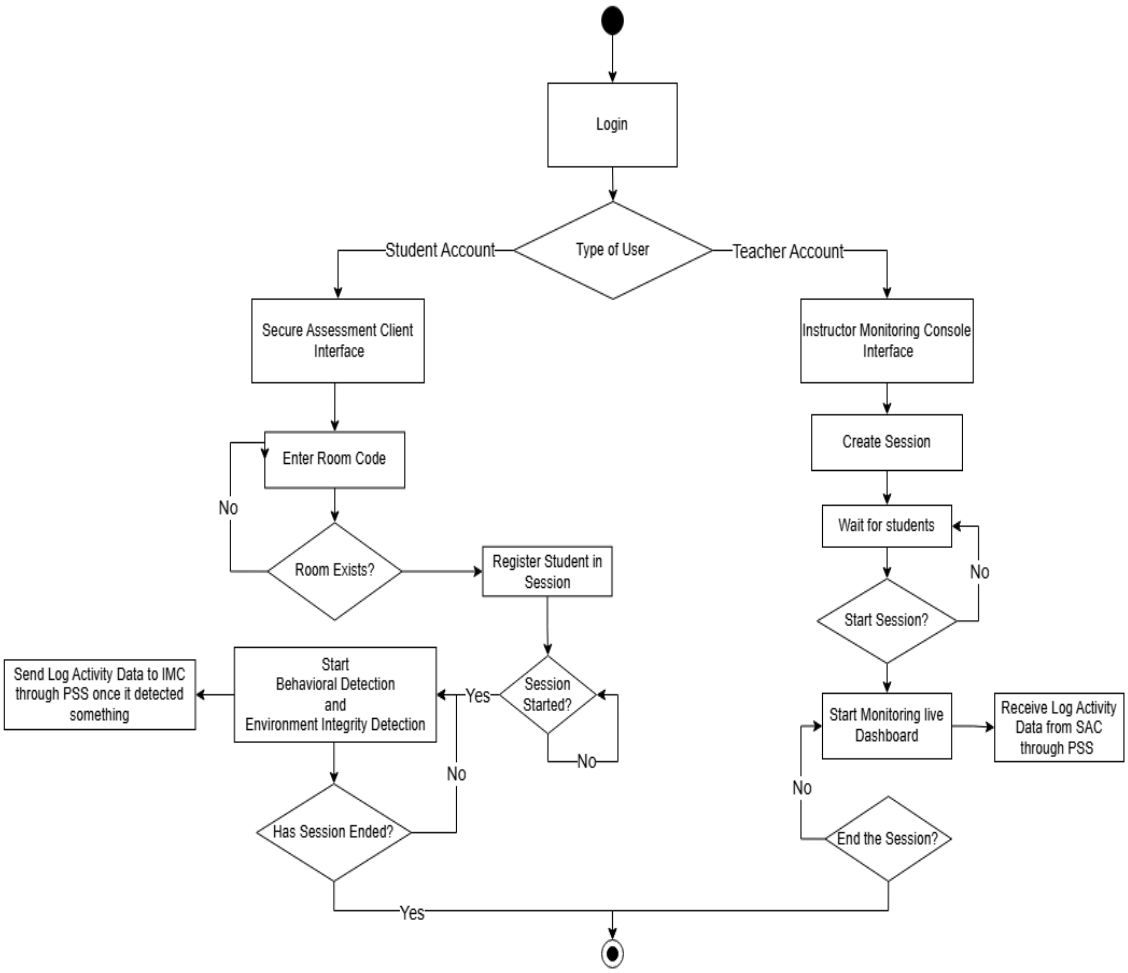

Figure 2. Activity Diagram

Figure 2 shows the system's workflow, which begins with the user logging in and being identified as either a student or a teacher. If the user is a teacher, they access the Instructor Monitoring Console (IMC) to create a room and wait for students to connect. Once the session starts, the teacher views a live dashboard that updates in real-time as data is received. On the other hand, students use the Secure Assessment Client (SAC) to enter a room code and join the session. When the exam begins, the app runs locally to

monitor violations, such as unauthorized apps or changing windows. If any suspicious activity is detected, the logs are sent through the Proctoring Session Server (PSS) to the teacher's dashboard, continuing until the session is ended

3.2.2 Use Case Diagram

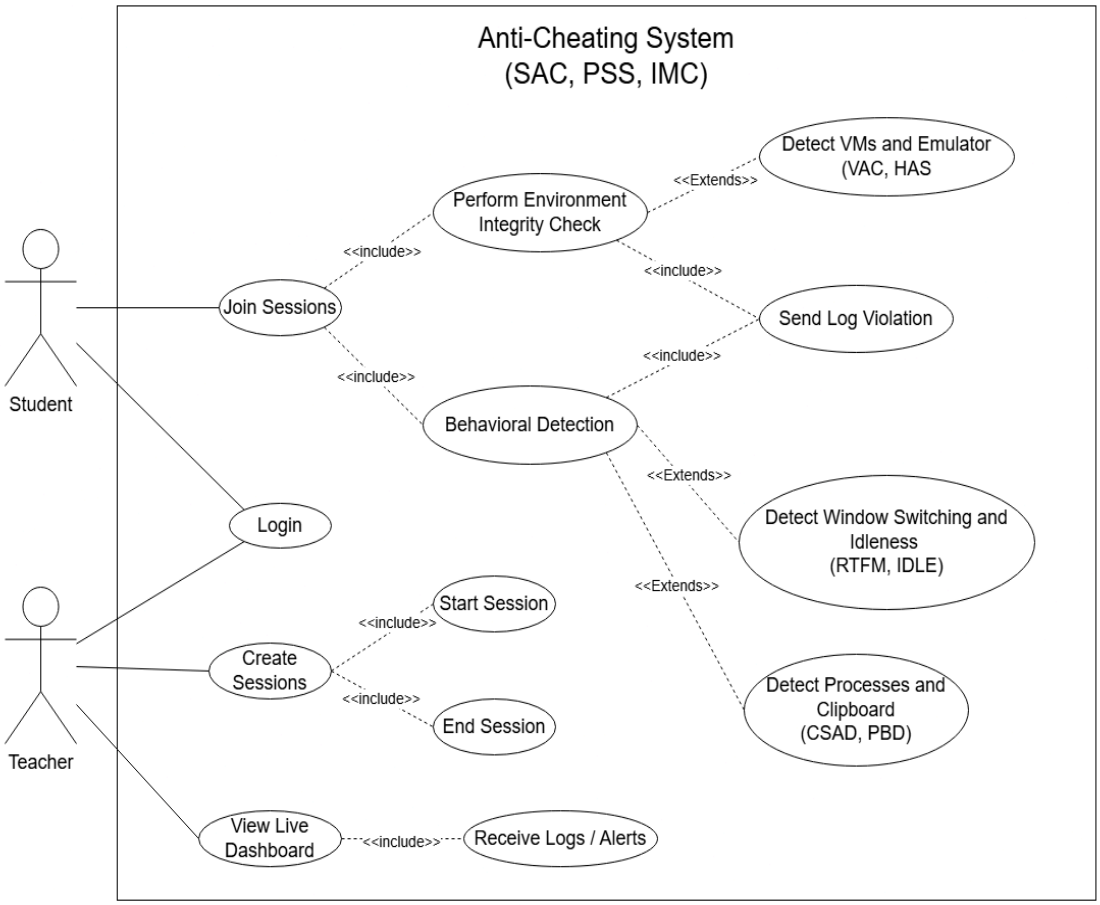

Figure 3. Use Case Diagram

Figure 3 shows the interaction between the users and the Anti-Cheating System. The student connects to the system to Join a Session, which automatically triggers the mandatory Environment Integrity Checks and Behavioral Detection. These background processes continuously validate the student's actions and automatically send log

violations to the server if a rule is broken. On the other side, the Teacher acts as the session manager. They have the authority to Create, Start, and End Sessions. During an active exam, the teacher utilizes the View Live Dashboard use case, which includes receiving real-time alerts from the students, allowing for immediate intervention if cheating is detected.

3.2.2 Class Diagram

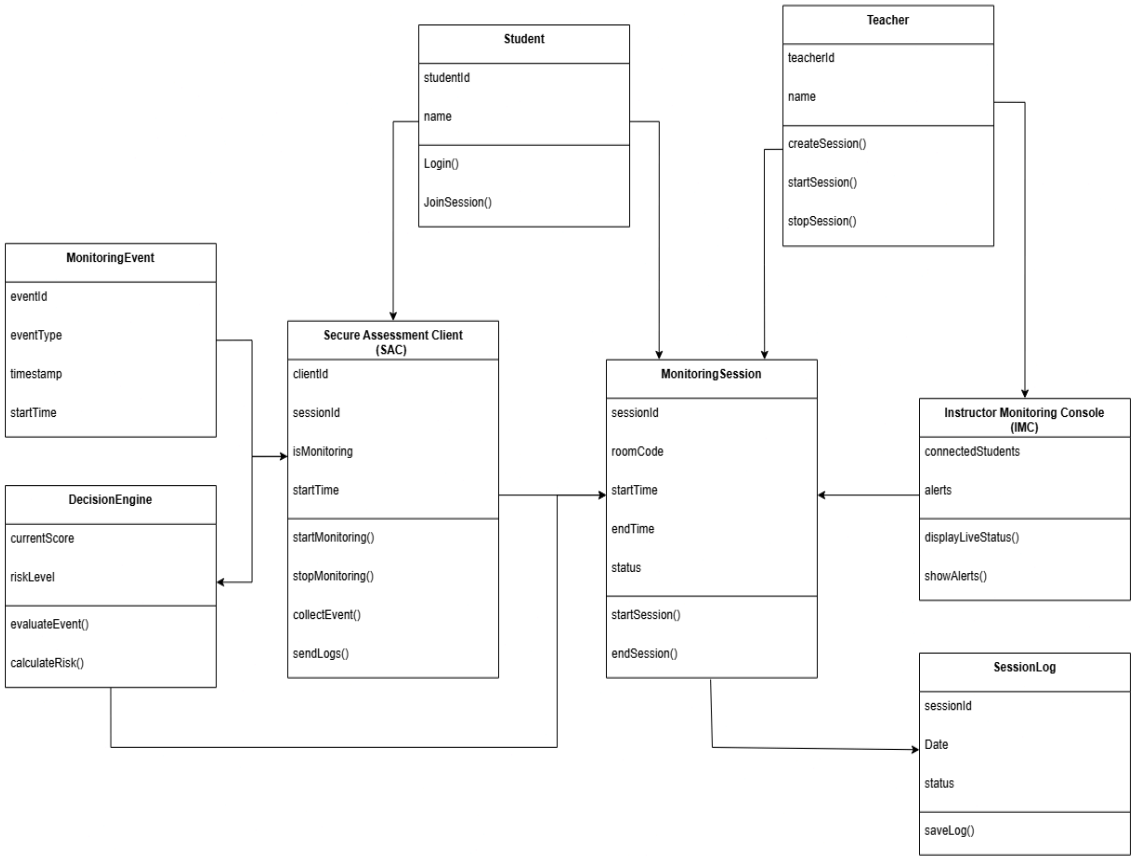

Figure 4. Class Diagram

Figure 4 shows the data structure and relationships between the system's components. The Student and Teacher classes represent the primary users, each having unique roles. Students join sessions via the Secure Assessment Client (SAC), while teachers manage

exams using the Instructor Monitoring Console (IMC). The MonitoringSession acts as the central link, connecting students to a specific room code created by the teacher. During an active session, the SAC generates MonitoringEvent objects (such as Alt-Tab detection or IDLE status). These events are processed by the DecisionEngine, which calculates a risk level (Safe, Suspicious, or Cheating) before sending the data to the server. Finally, all session activities are permanently recorded in the SessionLog, ensuring that a complete history of the exam is saved for the teacher's review.

3.3 System Architecture

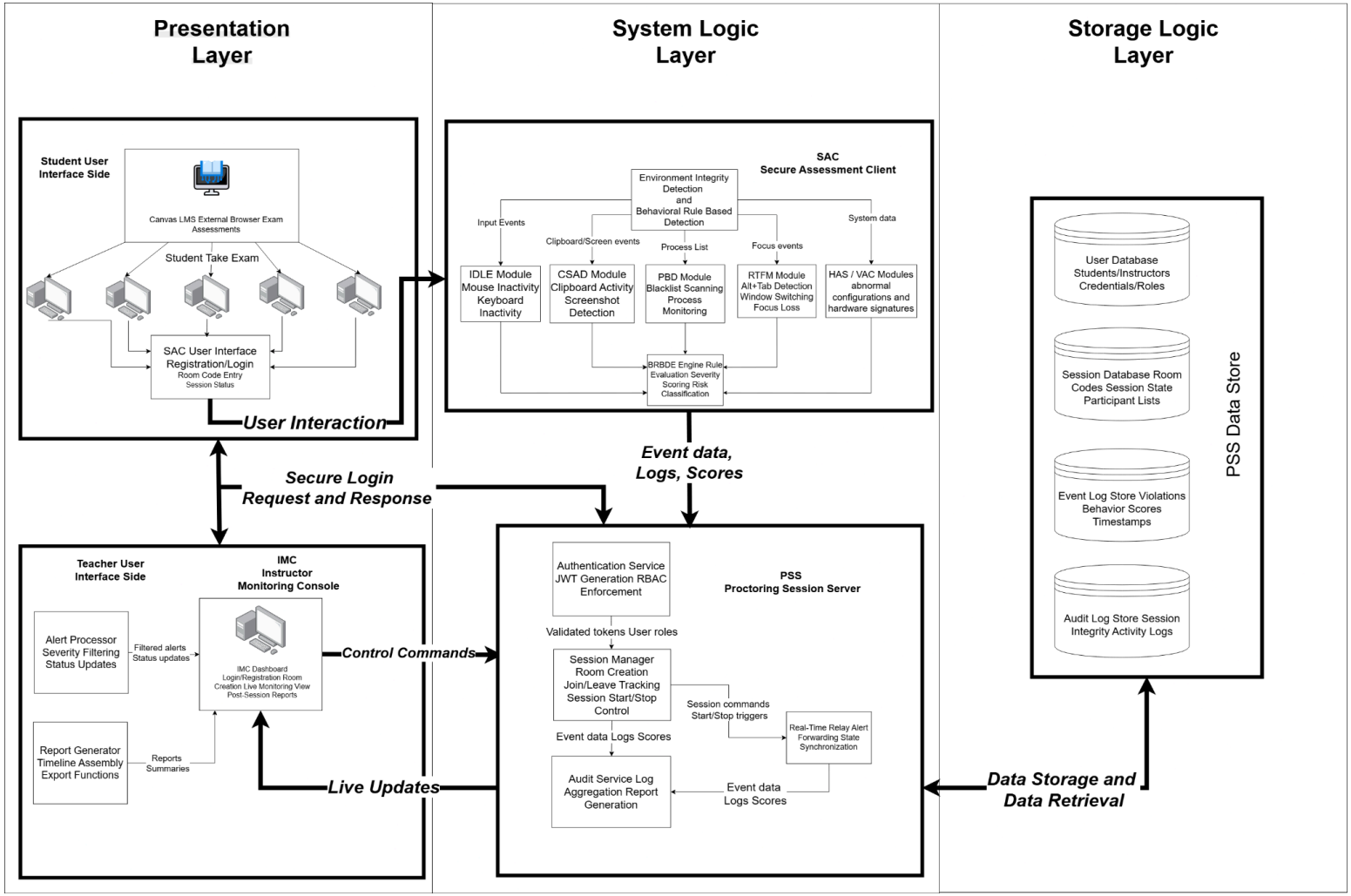

Figure 5. System Architecture

Figure 5. System Architecture illustrates the three-layered design of the proposed secure online assessment monitoring system, consisting of the Presentation Layer, System Logic Layer, and Storage Logic Layer. The Presentation Layer provides separate interfaces for students and instructors, where students complete their exams through an external Canvas LMS while the Secure Assessment Client (SAC) operates in parallel to manage registration, session participation, and status display, and instructors use the Instructor Monitoring Console (IMC) to control monitoring sessions, view live alerts, and access post-session reports. The System Logic Layer contains the core processing components, including SAC detection modules and the Behavioral Rule-Based Decision Engine (BRBDE), which evaluates monitored activities and generates behavior scores and risk classifications. These results are coordinated by the Proctoring Session Server (PSS), which handles authentication, session management, real-time alert distribution, and audit logging. The Storage Logic Layer securely maintains user credentials, session information, monitoring events, and audit records. This architectural design supports scalability, reliability, and ease of maintenance through clear separation of responsibilities and centralized session coordination.

3.4 Hardware and Software Requirements

The development and deployment of the Multi-Layered Anti-Cheating System require specific hardware and software configurations to ensure the stability of the application level monitoring and the secure relay of data between the student and instructor components. The system is designed to run efficiently on standard university laboratory computers and personal laptops without needing high end gaming hardware.

3.4.1 Hardware Requirements

The system is intended to operate smoothly on consumer grade desktop or laptop computers. Both participants and hosts use similar hardware since the monitoring and viewing processes run simultaneously with an external assessment platform such as Canvas through a web browser. Table 1 presents the minimum and recommended hardware specifications needed to support the Secure Assessment Client, and the Instructor Monitoring Console.

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 or AMD Ryzen 3	Intel Core i5 or AMD Ryzen 5
Memory	4 GB RAM	8 GB RAM or higher
Storage	5 GB available disk space	Solid State Drive recommended
Display	1366 x 768 resolution	1920 x 1080 resolution
Network	Stable LAN or controlled local network	Wired or high speed LAN connection

Table 1. Hardware Requirements

3.4.2 Software Requirements

The software environment is strictly limited to the Windows platform to maintain consistent behavior and monitoring accuracy. The system operates entirely at the application level and does not require kernel level drivers or administrative system modifications. Table 2 summarizes the software requirements used for development and deployment.

Component	Specification
Operating System	Windows 10 or later
Programming Language	C#
Framework	.NET 10
User Interface	WPF (Windows Presentation Foundation)
Server Framework	ASP.NET Core
Real Time Communication	SignalR
Database	MySQL
Security	HTTPS and JWT authentication
Integrated development environment	Visual Studio 2026

Table 2. Software requirements

3.5 Methods in Developing Software

The Multi-Layered Anti-Cheating Desktop Application was developed using a systematic and iterative software development process aligned with the applied research approach of the study. Since applied research emphasizes the creation and evaluation of practical solutions to real-world problems, the development process focused on translating

identified requirements into a functional system suitable for actual use in online assessment environments.

Development began with requirements analysis, where functional and non-functional requirements were identified based on the study objectives, challenges observed in existing online assessment systems, and findings from related literature. This phase ensured that the system design directly addressed issues related to academic integrity, usability, and user trust.

Following requirement analysis, system design was conducted to define the overall architecture, system modules, data flow, and interactions among the Secure Assessment Client (SAC), Proctoring Session Server (PSS), and Instructor Monitoring Console (IMC). The design emphasized modularity and scalability to support continuous monitoring and real-time communication during examination sessions.

The implementation phase involved developing core system components, including environment integrity checks, behavioral rule-based detection mechanisms, decision-making logic, and real-time logging features. Development was performed incrementally, allowing continuous testing and refinement of features to ensure system accuracy, security, and performance. Feedback gathered during development and testing phases was used to improve system functionality until the application met its intended objectives for secure online assessment.

3.5.1 Agile Methodology

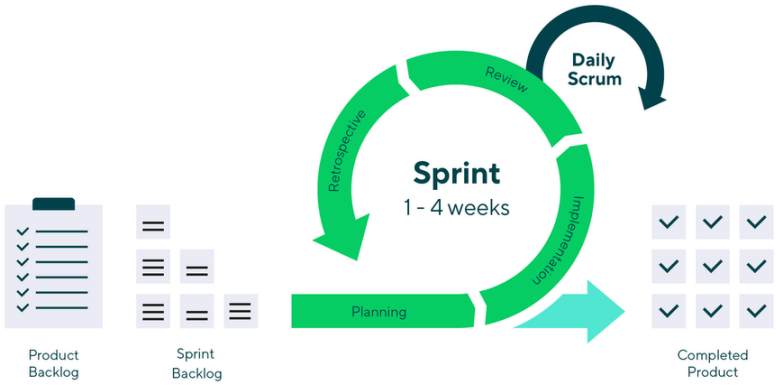

Figure 6. Agile Methodology from Wrike

The development of the Anti-Cheating Desktop Application followed the Agile software development methodology to support flexibility and continuous improvement throughout the project lifecycle. Agile was selected because it allows iterative development, frequent testing, and timely adaptation to changes—an essential characteristic for security-focused systems where requirements may evolve based on testing outcomes and emerging cheating behaviors.

The system was developed in incremental iterations, with each cycle focusing on specific features such as environment integrity checks, behavioral monitoring, rule-based decision logic, and real-time logging. Each iteration included planning, development, testing, and review phases to ensure that implemented features met functional and performance requirements before proceeding to the next cycle.

This iterative approach enabled early identification of system issues and allowed refinements based on testing results and user feedback. By applying Agile methodology, the development process remained aligned with the applied research objective of producing a practical, usable, and reliable system for real-world online assessment environments.

3.5.2 Algorithm

The proposed system uses a sequence of connected algorithms to monitor the assessment environment, observe participant behavior, and determine the likelihood of cheating. These algorithms work together in a layered manner, where the output of one process becomes the input of the next.

1.) Environment Integrity Detection Algorithm

The process begins with the Environment Integrity Detection Algorithm (EIDA). This algorithm is executed when a participant joins a monitoring session. Its purpose is to verify whether the computer environment is legitimate and suitable for secure monitoring. The algorithm checks virtualization artifacts by scanning system files, drivers, and hardware identifiers associated with virtual machines and emulators.

```
NAME
    Environment Integrity Detection Algorithm
DOCUMENTATION
    This algorithm verifies whether the student system is running in a trusted environment.
PURPOSE
    DETECT virtualization and abnormal system configurations.
CONTEXT
    FACT monitoring_session_active = true;
    FACT environment_status = "Normal";
BODY
    ASSIGN system_info = collect_system_information;
    ASSIGN hardware_info = collect_hardware_identifiers;
    ASSIGN service_list = collect_running_services;

    IF virtualization_artifacts_detected(system_info, service_list) THEN
        ASSIGN environment_status = "Virtualized";
        RECORD violation "Virtual Machine Detected";
    ENDIF

    IF emulator_signatures_detected(system_info) THEN
        ASSIGN environment_status = "Virtualized";
        RECORD violation "Android Emulator Detected";
    ENDIF

    IF abnormal_configuration_detected(hardware_info) THEN
        ASSIGN environment_status = "Suspicious";
        RECORD violation "Abnormal Hardware Software Configuration";
    ENDIF

    EXECUTE send_environment_status environment_status;
END
```

Figure 7. Pseudocode of Environment Integrity Detection Algorithm

2.) Behavioral Rule Based Detection Algorithm

Once the environment is verified, the system activates the Behavioral Rule Based Detection Algorithm (BRBDA). This algorithm continuously monitors participant actions during the assessment period. It observes window focus changes through real time focus monitoring, detects unauthorized applications through process blacklisting detection, and tracks copy paste and screenshot attempts using clipboard and screenshot activity detection. Each detected action is treated as a behavioral event and logged with a timestamp.

```
NAME
    Behavioral Rule Based Detection Algorithm
DOCUMENTATION
    This algorithm monitors user actions and records suspicious behavior during the exam.
PURPOSE
    DETECT rule based behavioral violations.
CONTEXT
    FACT monitoring_session_active = true;
    FACT detected_events = empty_list;
BODY
    WHILE monitoring_session_active DO
        IF window_focus_lost THEN
            RECORD detected_events "Window Switch Detected";
        ENDIF

        IF forbidden_process_running THEN
            RECORD detected_events "Unauthorized Application Detected";
        ENDIF

        IF clipboard_activity_detected THEN
            RECORD detected_events "Copy Paste Activity Detected";
        ENDIF

        IF screenshot_key_pressed THEN
            RECORD detected_events "Screenshot Activity Detected";
        ENDIF

        IF keyboard_mouse_idle_exceeded THEN
            RECORD detected_events "Idle Behavior Detected";
        ENDIF
    ENDWHILE

    EXECUTE send_detected_events detected_events;
END
```

Figure 8. Pseudocode of Behavioral Rule Based Detection Algorithm

3.) Behavioral Rule Based Detection Algorithm

The collected environment integrity results and behavioral event logs are then analyzed by the Behavioral Rule Based Decision Engine (BRBDE). This algorithm evaluates the combined data using predefined rules. Each detected event is assigned a severity score based on its risk level, and repeated or combined violations increase the overall score. The decision engine analyzes behavior patterns across the entire session rather than relying on single events.

Based on the total severity score and rule evaluation, the system classifies the participant's status as safe, suspicious, or cheating.

```
NAME
    Behavioral Rule Based Decision Engine
DOCUMENTATION
    This algorithm evaluates detected behaviors and determines the cheating risk level.
PURPOSE
    CLASSIFY student behavior based on accumulated rule violations.
CONTEXT
    FACT total_score = 0;
    FACT risk_level = "Safe";
BODY
    FOR EACH event IN detected_events DO
        IF event == "Window Switch Detected" THEN
            ASSIGN total_score = total_score + 1;
        ENDIF

        IF event == "Unauthorized Application Detected" THEN
            ASSIGN total_score = total_score + 3;
        ENDIF

        IF event == "Copy Paste Activity Detected" THEN
            ASSIGN total_score = total_score + 2;
        ENDIF

        IF event == "Screenshot Activity Detected" THEN
            ASSIGN total_score = total_score + 2;
        ENDIF

        IF event == "Idle Behavior Detected" THEN
            ASSIGN total_score = total_score + 1;
        ENDIF
    ENDFOR

    IF total_score < 3 THEN
        ASSIGN risk_level = "Safe";
    ELSE IF total_score BETWEEN 3 AND 5 THEN
        ASSIGN risk_level = "Suspicious";
    ELSE
        ASSIGN risk_level = "Cheating";
    ENDIF

    EXECUTE send_risk_result risk_level;
END
```

Figure 9. **Pseudocode of Behavioral Rule Based Decision Engine**

3.5.3 Log Structure and Tagging Mechanism

To ensure structured behavioral assessment and environmental integrity validation, the system monitors specific, measurable operating system and user-behavior parameters during the examination session. All detected activities are transformed into standardized, tagged log objects before transmission to the Proctoring Session Server (PSS).

The monitoring framework is divided into two primary domains:

1. Environment Integrity Monitoring
2. Behavioral Activity Monitoring

1. Environment Integrity Monitoring

Environment monitoring verifies whether the examination session is executed in a secure and uncompromised computing environment.

1.1 Virtualization Artifact Check (VAC)

Monitored Indicators:

- Presence of virtual machine drivers
- VM-specific registry keys
- Hypervisor process names
- Virtual network adapters
- Suspicious BIOS strings

Trigger Condition 1: Detection of ≥ 1 confirmed virtualization artifact.

Generated Tag: ENV_VAC Violation Type: Aggressive Purpose: Prevents execution of the exam inside sandboxed or emulated environments that may allow bypassing monitoring controls.

1.2 Hardware-Software Artifact Scan (HAS)

Monitored Indicators:

- Remote desktop services activation
- Debugging tools running
- Screen recording services
- Macro automation tools
- System time manipulation attempts

Trigger Condition 1: Detection of unauthorized system-level assistance tools.

Generated Tag: ENV_HAS Violation Type: Aggressive Purpose: Ensures no secondary control systems interfere with the exam session.

2. Behavioral Activity Monitoring

Behavioral monitoring evaluates real-time student interaction patterns during the exam.

2.1 Real-Time Focus Monitoring (RTFM)

Monitored Indicators:

- Window focus loss events
- Application switching frequency
- Duration outside exam window
- Multi-monitor switching
- Browser tab changes (if embedded mode)

Trigger Conditions:

- Focus lost > 3 times within 60 seconds
- Focus loss duration > 10 seconds
- Repetitive window switching pattern

Generated Tag: BEH_RTFM Violation Type: Passive → escalates to Aggressive if repeated Purpose: Detects potential answer searching or collaboration attempts.

2.2 Process Blacklisting Detection (PBD)

Monitored Indicators:

- Launch of restricted applications:
    - Messaging apps
    - Remote access tools
    - AI chat platforms
    - Screen capture software
- Execution of known cheating utilities

Trigger Condition: Blacklisted process detected during session runtime.

Generated Tag: BEH_PBD Violation Type: Aggressive Purpose: Prevents parallel use of assistance software during examination.

2.3 Clipboard and Screenshot Activity Detection (CSAD)

Monitored Indicators:

- Copy command (Ctrl+C)
- Paste command (Ctrl+V)
- Clipboard buffer modification
- Screenshot key activation (PrintScreen, Snipping Tool)

Trigger Conditions:

- Clipboard event while exam window active
- Screenshot event during session

Generated Tag: BEH_CSAD Violation Type: Passive (single) → Aggressive (repeated) Purpose: Prevents unauthorized content extraction or insertion.

3. Structured Log Object Format

Field Name	Description
LogID	Unique identifier for each event
SessionID	Unique exam session reference
UserID	Student identifier
PlatformTag	Source platform classification
MonitoringDomain	ENV or BEH
EventType	Specific detection module (VAC, HAS, RTFM, PBD, CSAD)
TriggerMetric	Measured indicator value
Threshold	Configured rule limit
ViolationType	Passive or Aggressive
SeverityScore	PassiveNumerical risk score assigned by BRBDEor Aggressive
Timestamp	Date and time of occurrence
DeviceHash	Hardware fingerprint identifier
Status	Safe / Suspicious / Cheating

Table 3. Structured Log Object Format

4. Severity Escalation Logic

The Behavioral Rule-Based Decision Engine (BRBDE) assigns severity levels:

Severity Level	Description
S1	Single Passive Event
S2	Repeated Passive Event
S3	Aggressive Violation
S4	Combined Critical Behavior

Table 4: BRBDE Assigns Severity Levels

Escalation Rules:

- 3 Passive events → Upgraded to S2
- 1 Aggressive event → Minimum S3
- Mixed ENV + BEH violations → S4

5. Example Log Layout

```
LogID: 2026-03-001245
SessionID: EXAM-IT2026-01
UserID: STU-034
PlatformTag: ExternalBrowserMode
EventCategory: BEH
EventType: BEH_RTFM
ViolationType: Aggressive
SeverityScore: 3
Timestamp: 2026-03-05 10:24:32
DeviceHash: A94F3B12X
Status: Suspicious
```

3.5.4 Detection Effectiveness Measurement Process

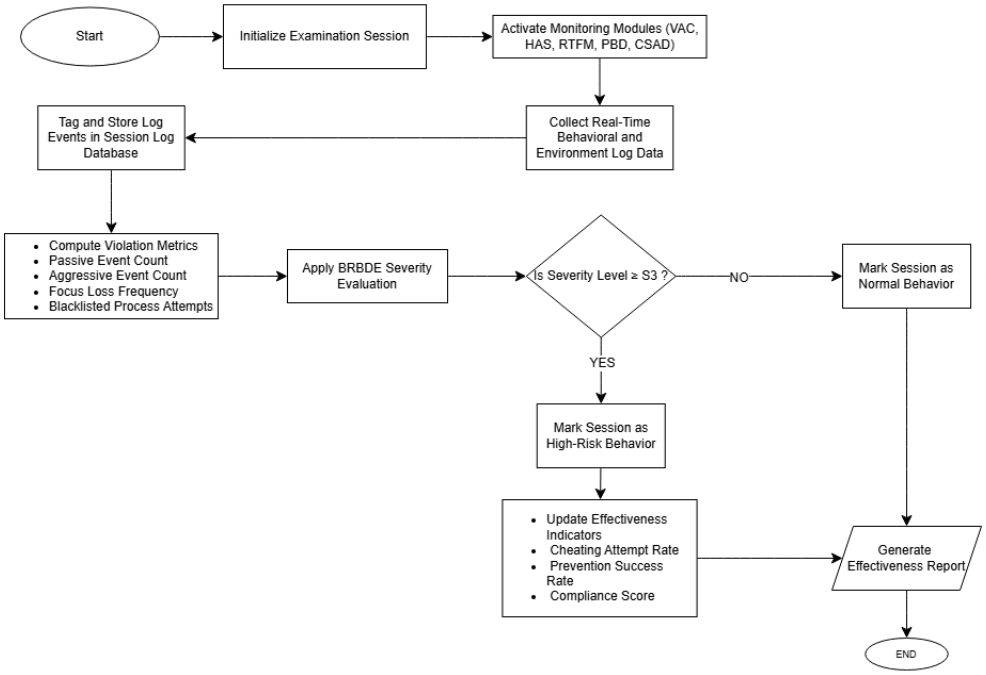

Figure 10. **Flowchart of Effectiveness Measurement Process**

The system measures its detection effectiveness by analyzing logged behavioral and environmental events during the examination session. Once the session begins, all monitoring modules are activated and real-time events are collected and stored. Each event is tagged and classified as passive or aggressive. The system computes violation metrics such as event counts, focus loss frequency, and blacklisted process attempts. These values are evaluated by the Behavioral Rule-Based Decision Engine (BRBDE), which assigns a severity level. If the severity reaches S3 or higher, the session is marked as high-risk; otherwise, it is considered normal. The computed results are then summarized in the effectiveness report, showing how accurately the system detected suspicious behavior.

3.5.5 Security and Integrity Enforcement Process During Online Assessment

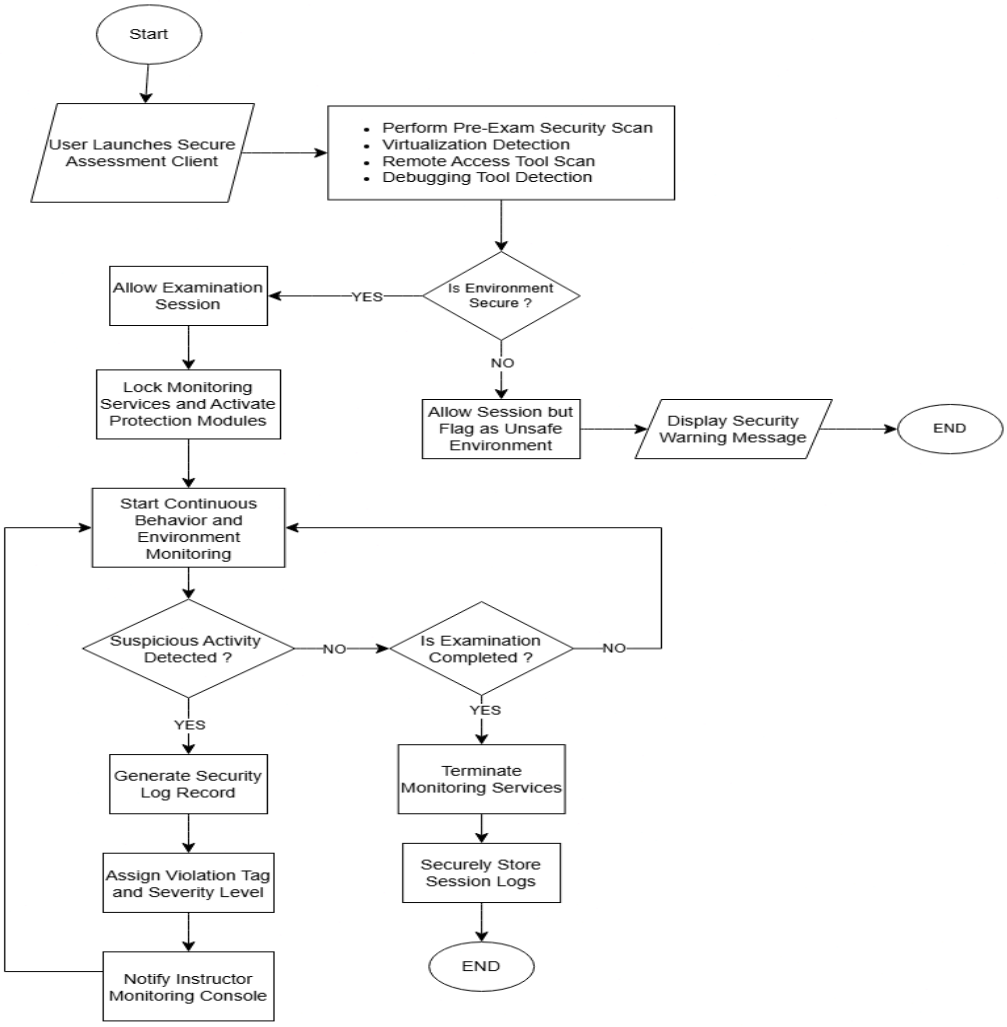

Figure 11. **Detection Effectiveness Measurement Process**

To ensure exam security, the system performs a pre-exam validation and continuous monitoring process.When the Secure Assessment Client (SAC) is launched, it scans for virtualization, remote access tools, debugging tools, and unsafe configurations. If the environment is secure, the exam proceeds and monitoring services are activated. If unsafe, the session is flagged and a warning is displayed. During the exam, the system

continuously monitors focus changes, running processes, clipboard usage, and screenshot attempts. Suspicious activities are logged, assigned a severity level, and reported to the Instructor Monitoring Console (IMC). Monitoring ends when the exam is completed, and all logs are securely stored to maintain session integrity.

3.6 Methods in Evaluating the System

The Anti-Cheating Desktop Application was evaluated to determine its effectiveness, usability, performance, reliability, and user trust in the context of online examinations. The evaluation approach is consistent with the applied, descriptive comparative research design of the study, focusing on assessing user perceptions and system behavior after actual system use rather than experimental manipulation.

System effectiveness was evaluated by examining the system's ability to detect cheating-related behaviors, including unauthorized application usage, window switching, virtualization artifacts, and idle activity. Controlled testing scenarios were conducted, and system-generated logs were reviewed to determine detection accuracy, consistency, and reliability. These logs served as objective evidence of the system's monitoring capability during assessment sessions.

System usability was assessed using the System Usability Scale (SUS), which gathered quantitative feedback from both students and instructors after using the system. The SUS instrument measured perceived ease of use, learnability, and overall satisfaction, allowing

comparison of usability perceptions between traditional browser-based online assessments and assessments integrated with the anti-cheating desktop application.

System performance was evaluated by observing responsiveness and resource utilization during active examination sessions. This evaluation ensured that the monitoring processes operated efficiently and did not interfere with normal assessment activities or negatively affect user experience.

System reliability and auditability were evaluated through analysis of session logs to verify the completeness, accuracy, and consistency of recorded monitoring events. These logs ensured that all detected behaviors and alerts were properly documented and retrievable for instructor review.

Overall evaluation results were analyzed using descriptive statistical methods and aligned with recognized software quality characteristics, including functional suitability, usability, performance efficiency, reliability, and security, to support a comprehensive assessment of the system's effectiveness and user acceptance.

3.7 Testing Methods

3.7.1 Alpha Testing

Alpha testing will be conducted by the development team during the early stages of implementation to verify the core features of the system such as authentication, session handling, environment integrity detection, behavioral monitoring, and real time logging.

3.7.2 White box Testing

White box testing will be used to examine the internal logic of the system. This includes validating rule conditions, detection logic, and score computation to ensure correct program flow.

3.7.3 Integration Testing

Integration testing will be performed to verify proper communication between the Secure Assessment Client, Proctoring Session Server, and Instructor Monitoring Console.

3.7.4 Beta Testing

Beta testing is planned to be conducted after the core features of the system have passed internal testing. This phase involves a limited group of actual users, specifically selected students and instructors, who will use the system in a controlled environment.

3.8 Data Gathering Procedures

Data gathering for the study employed quantitative methods consistent with an applied and descriptive comprehensive research design. Data collection focused on evaluating system effectiveness, usability, performance, and user trust under two assessment conditions: (1) traditional browser-based online assessment and (2) online assessment integrated with the anti-cheating desktop application.

During the alpha testing phase, system-generated data were collected by the developers. These data included monitor logs, detected violations, system response times, and error

reports. The collected information served as objective evidence for evaluating detection accuracy, system reliability, and functional correctness.

During the **beta testing phase**, data were gathered from selected student and instructor respondents who utilized the system during actual assessment sessions. System-generated logs recorded behavioral violations, session activities, and alert notifications. These logs were used to assess the system's effectiveness and reliability in a real-world setting.

Usability and user perception data were collected using standardized Likert-scale questionnaires, including the **System Usability Scale (SUS)**. The questionnaires were administered to respondents after completing assessment sessions under both assessment conditions. Performance-related data, such as system responsiveness and resource utilization, were also recorded during active sessions.

All collected data were consolidated and prepared for descriptive statistical analysis to compare user perceptions and system performance across the two assessment environments.

3.9 Respondents of the Study and Sampling Techniques

The respondents of this study consisted of one hundred five (105) participants from the Bachelor of Science in Computer Science and Software Engineering (BSCSSE) program and faculty members of FEU Tech Manila. The student respondents were composed of thirty-five (35) 1st year students, thirty-five (35) 2nd year students, and thirty (30) 3rd year students. In addition, five (5) faculty members participated in the evaluation. These

respondents were selected because they have experience with online examinations and are familiar with computer-based assessment environments.

The study used purposive sampling, meaning that participants were intentionally selected based on their relevance to the research. The chosen students and faculty members have prior exposure to online assessments, making them suitable individuals to evaluate the proposed Multi-Layered Anti-Cheating Desktop Application. This approach ensured that the feedback gathered reflects realistic academic use of the system.

All respondents were required to use the system before answering the System Usability Scale (SUS) questionnaire. Their responses were used to evaluate the system's usability, ease of use, and overall acceptance. The data collected from the 105 respondents served as the basis for analyzing user perceptions of the proposed system.

Table 3

Target Respondents

Respondent	No. Of Respondents
BSCSSE 1st year students	35
BSCSSE 2nd year students	35
BSCSSE 3rd year students	30
Faculty Members	5
Total Respondents	105

3.10 Statistical Treatment of Data

The data gathered in this study were analyzed using descriptive statistical methods to evaluate the usability, effectiveness, reliability, and user acceptance of the proposed Multi layered anti cheating desktop application. All quantitative data were obtained from system generated logs and responses to structured survey instruments, primarily the System Usability Scale (SUS) questionnaire administered to the respondents.

For the usability evaluation, SUS scores were computed following the standard SUS scoring procedure. Each item response was converted into a weighted score, summed, and multiplied by 2.5 to obtain a final SUS score ranging from 0 to 100. The weighted mean was used to determine the overall usability perception of the system across all respondents. This statistical measure was appropriate as it summarizes central tendencies in Likert-scale data and provides a clear interpretation of general user sentiment toward the system.

To measure the consistency and dispersion of responses, the standard deviation was calculated. This statistic provided insight into the level of agreement among participants regarding system usability and acceptance. A lower standard deviation indicated higher consensus among respondents, while a higher value suggested varied user experiences.

The reliability of the survey instrument was evaluated using Cronbach's alpha. This test was conducted to assess the internal consistency of the SUS questionnaire items. An

alpha coefficient of 0.70 or higher was considered acceptable, indicating that the instrument reliably measured the intended usability construct.

In addition to usability metrics, system performance and detection effectiveness were analyzed through frequency counts and percentage distributions derived from system logs. These statistics summarized the occurrence of detected behaviors such as window switching, unauthorized application usage, and environment integrity violations during controlled assessment sessions.

All statistical analyses were used to support the evaluation of the system based on ISO/IEC 25010 software quality characteristics, particularly usability, performance efficiency, reliability, and security. The results provided an objective basis for determining whether the proposed system met its design goals and was suitable for deployment in real-world online assessment environments.

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APPENDICES

APPENDIX A

Thesis Title Approval For

APPENDIX B

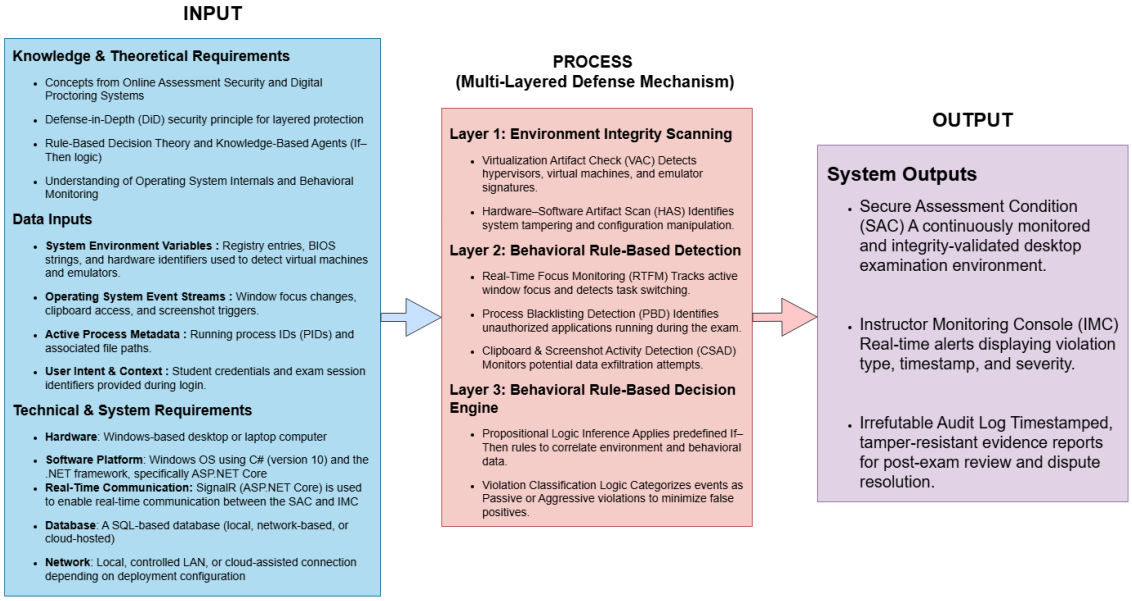


Figure B.1. *Input-Process-Output*

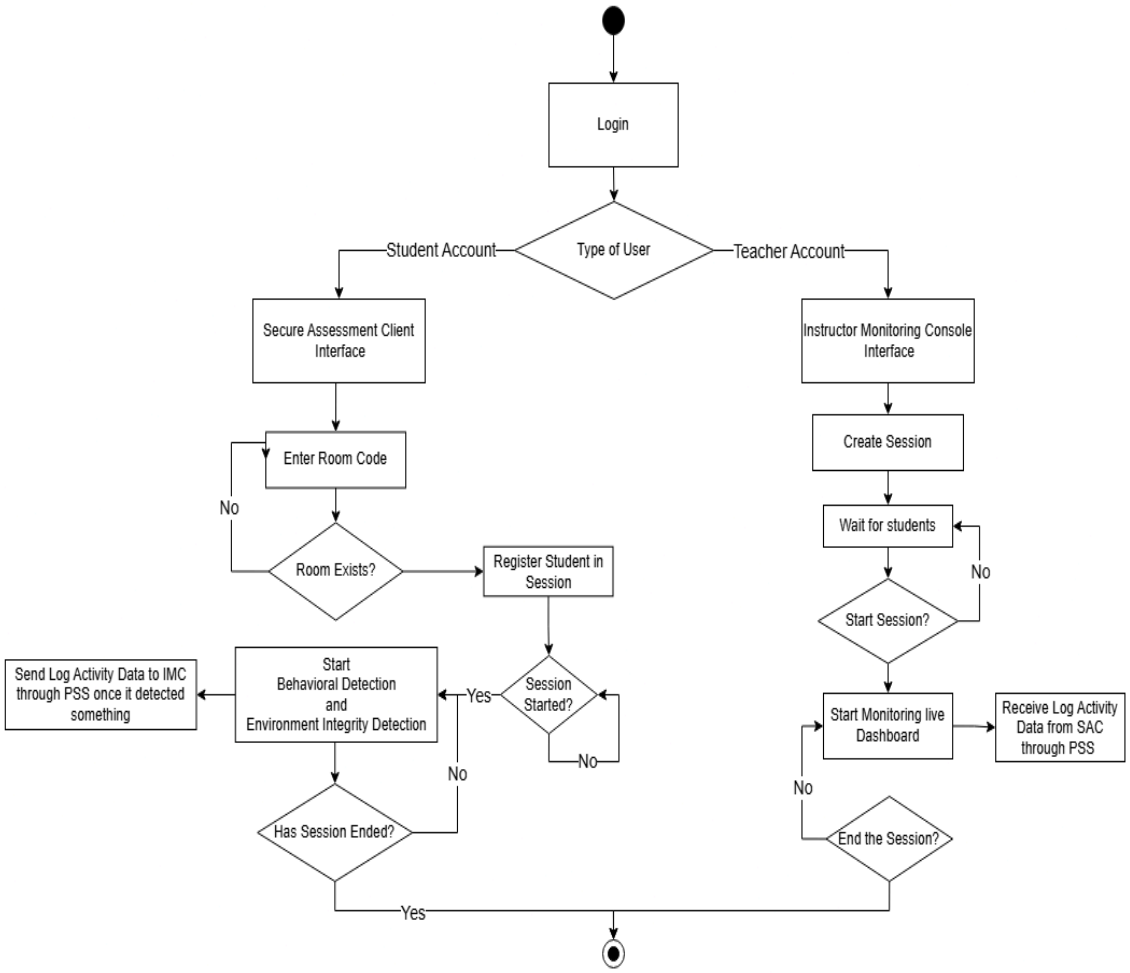


Figure B.2. Activity Diagram

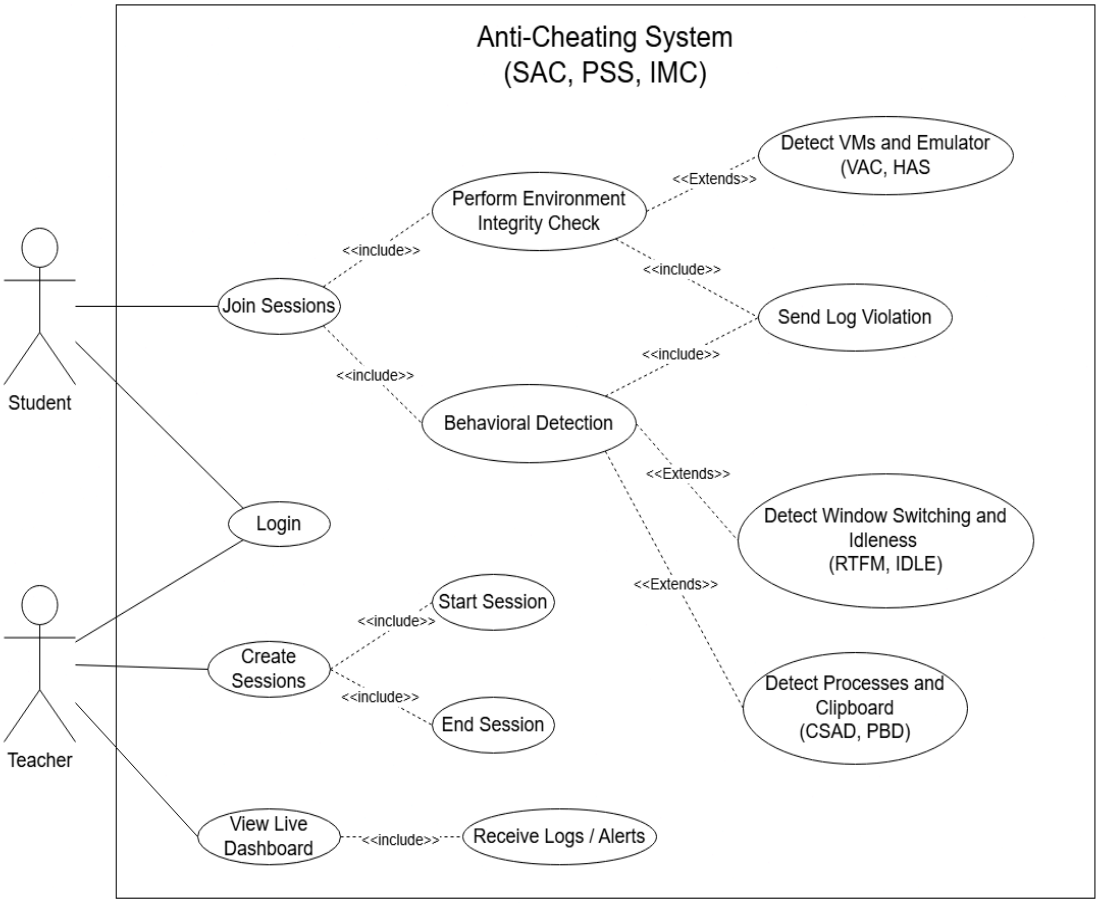
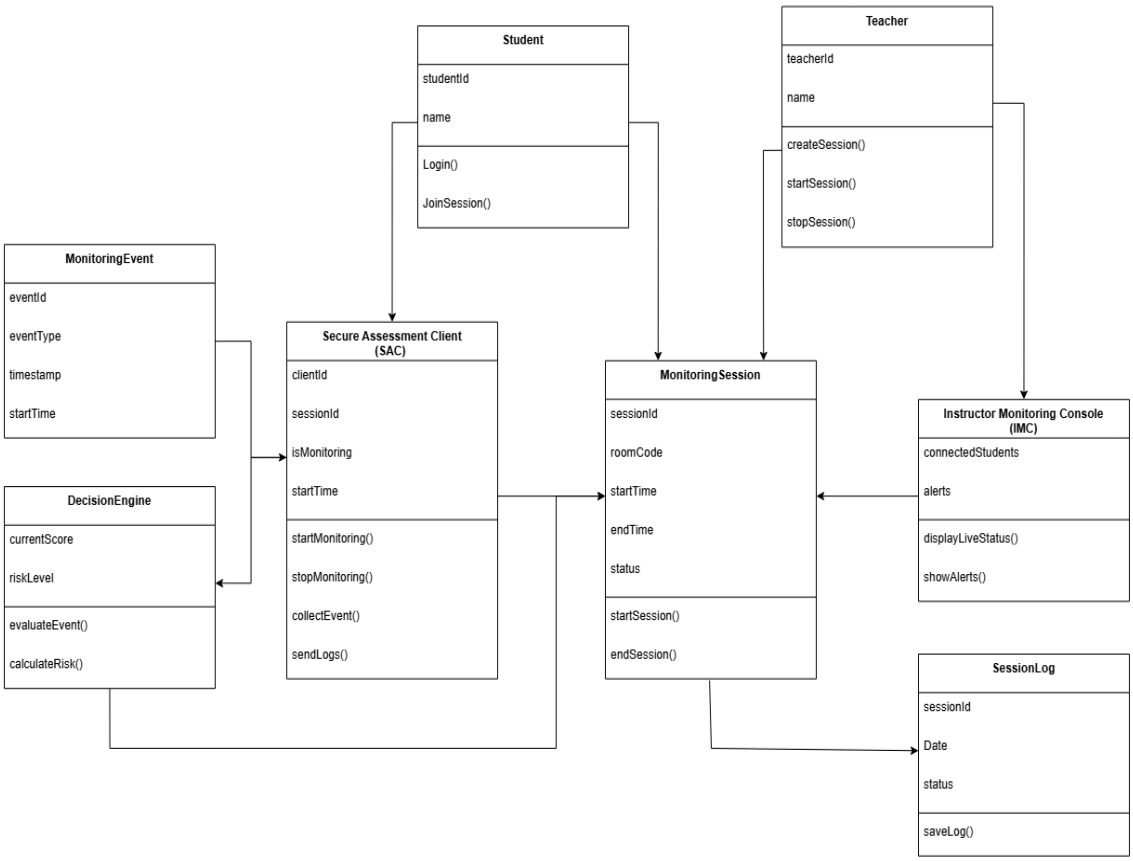
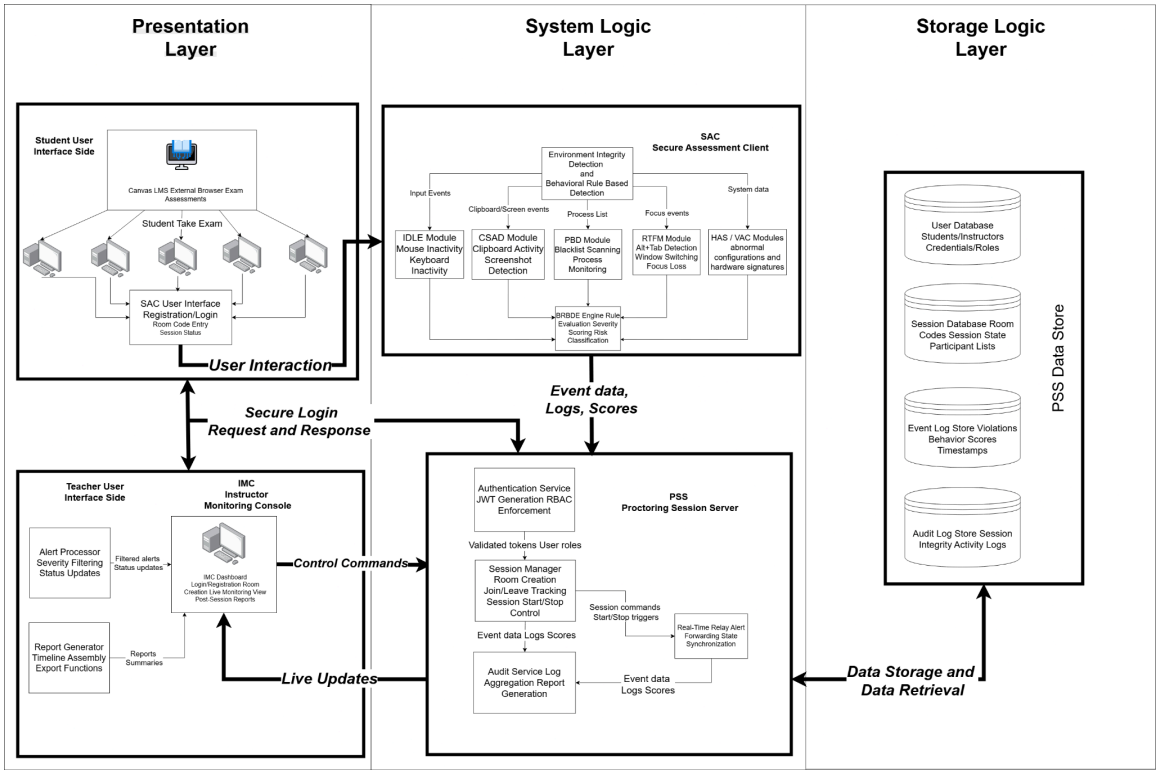


Figure B.3. Use Case Diagram



FigureB.4. Class Diagram



FigureB.5. System Architecture

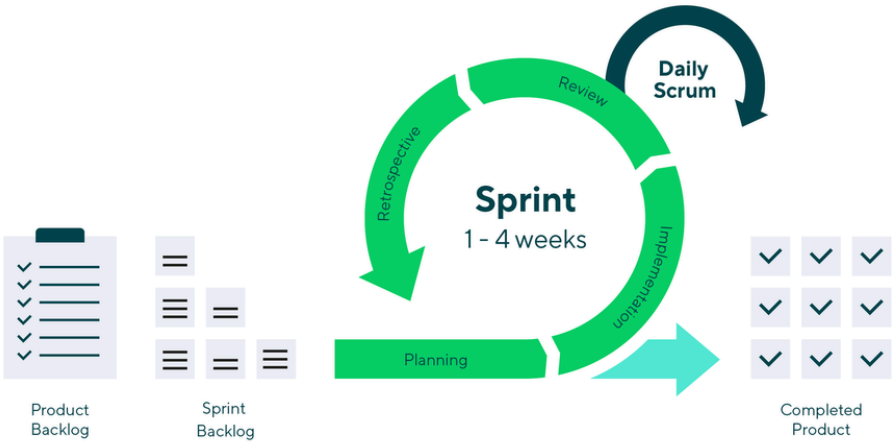


Figure B.6. Agile Methodology from Wrike

```
NAME
    Environment Integrity Detection Algorithm
DOCUMENTATION
    This algorithm verifies whether the student system is running in a trusted environment.
PURPOSE
    DETECT virtualization and abnormal system configurations.
CONTEXT
    FACT monitoring_session_active = true;
    FACT environment_status = "Normal";
BODY
    ASSIGN system_info = collect_system_information;
    ASSIGN hardware_info = collect_hardware_identifiers;
    ASSIGN service_list = collect_running_services;

    IF virtualization_artifacts_detected(system_info, service_list) THEN
        ASSIGN environment_status = "Virtualized";
        RECORD violation "Virtual Machine Detected";
    ENDIF

    IF emulator_signatures_detected(system_info) THEN
        ASSIGN environment_status = "Virtualized";
        RECORD violation "Android Emulator Detected";
    ENDIF

    IF abnormal_configuration_detected(hardware_info) THEN
        ASSIGN environment_status = "Suspicious";
        RECORD violation "Abnormal Hardware Software Configuration";
    ENDIF

    EXECUTE send_environment_status environment_status;
END
```

Figure B.7. Pseudocode of Environment Integrity Detection Algorithm

```
NAME
    Behavioral Rule Based Detection Algorithm
DOCUMENTATION
    This algorithm monitors user actions and records suspicious behavior during the exam.
PURPOSE
    DETECT rule based behavioral violations.
CONTEXT
    FACT monitoring_session_active = true;
    FACT detected_events = empty_list;
BODY
    WHILE monitoring_session_active DO
        IF window_focus_lost THEN
            RECORD detected_events "Window Switch Detected";
        ENDIF

        IF forbidden_process_running THEN
            RECORD detected_events "Unauthorized Application Detected";
        ENDIF

        IF clipboard_activity_detected THEN
            RECORD detected_events "Copy Paste Activity Detected";
        ENDIF

        IF screenshot_key_pressed THEN
            RECORD detected_events "Screenshot Activity Detected";
        ENDIF

        IF keyboard_mouse_idle_exceeded THEN
            RECORD detected_events "Idle Behavior Detected";
        ENDIF
    ENDWHILE

    EXECUTE send_detected_events detected_events;
END
```

Figure B.8. Pseudocode of Behavioral Rule Based Detection Algorithm

```
NAME
    Behavioral Rule Based Decision Engine
DOCUMENTATION
    This algorithm evaluates detected behaviors and determines the cheating risk level.
PURPOSE
    CLASSIFY student behavior based on accumulated rule violations.
CONTEXT
    FACT total_score = 0;
    FACT risk_level = "Safe";
BODY
    FOR EACH event IN detected_events DO
        IF event == "Window Switch Detected" THEN
            ASSIGN total_score = total_score + 1;
        ENDIF

        IF event == "Unauthorized Application Detected" THEN
            ASSIGN total_score = total_score + 3;
        ENDIF

        IF event == "Copy Paste Activity Detected" THEN
            ASSIGN total_score = total_score + 2;
        ENDIF

        IF event == "Screenshot Activity Detected" THEN
            ASSIGN total_score = total_score + 2;
        ENDIF

        IF event == "Idle Behavior Detected" THEN
            ASSIGN total_score = total_score + 1;
        ENDIF
    ENDFOR

    IF total_score < 3 THEN
        ASSIGN risk_level = "Safe";
    ELSE IF total_score BETWEEN 3 AND 5 THEN
        ASSIGN risk_level = "Suspicious";
    ELSE
        ASSIGN risk_level = "Cheating";
    ENDIF

    EXECUTE send_risk_result risk_level;
END
```

Figure B.9. Pseudocode of Behavioral Rule Based Decision Engine

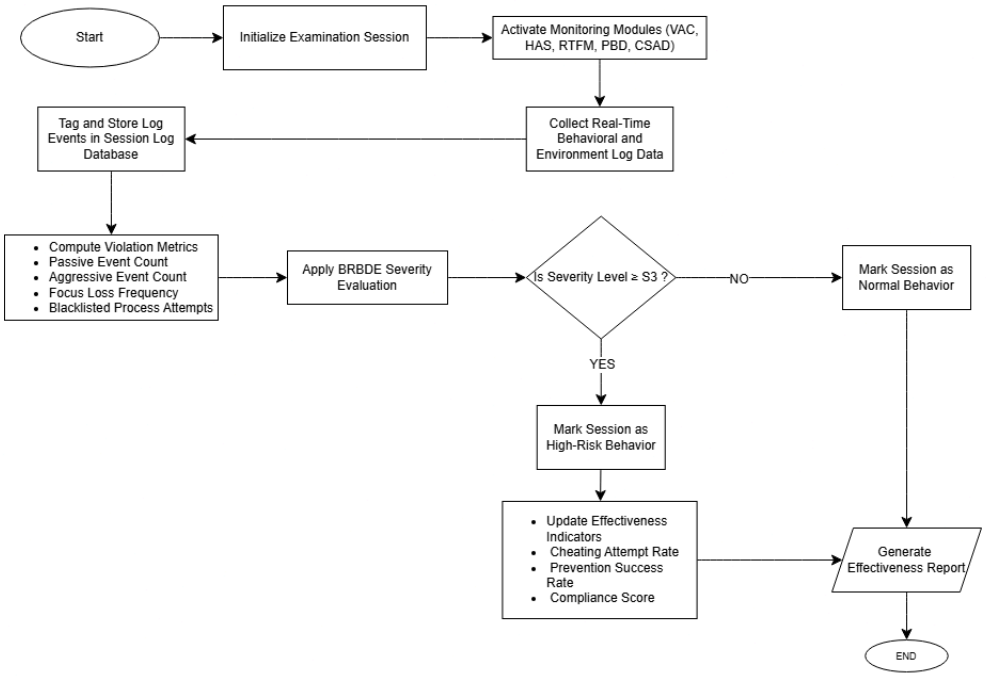


Figure B.10. Flowchart of Effectiveness Measurement Process

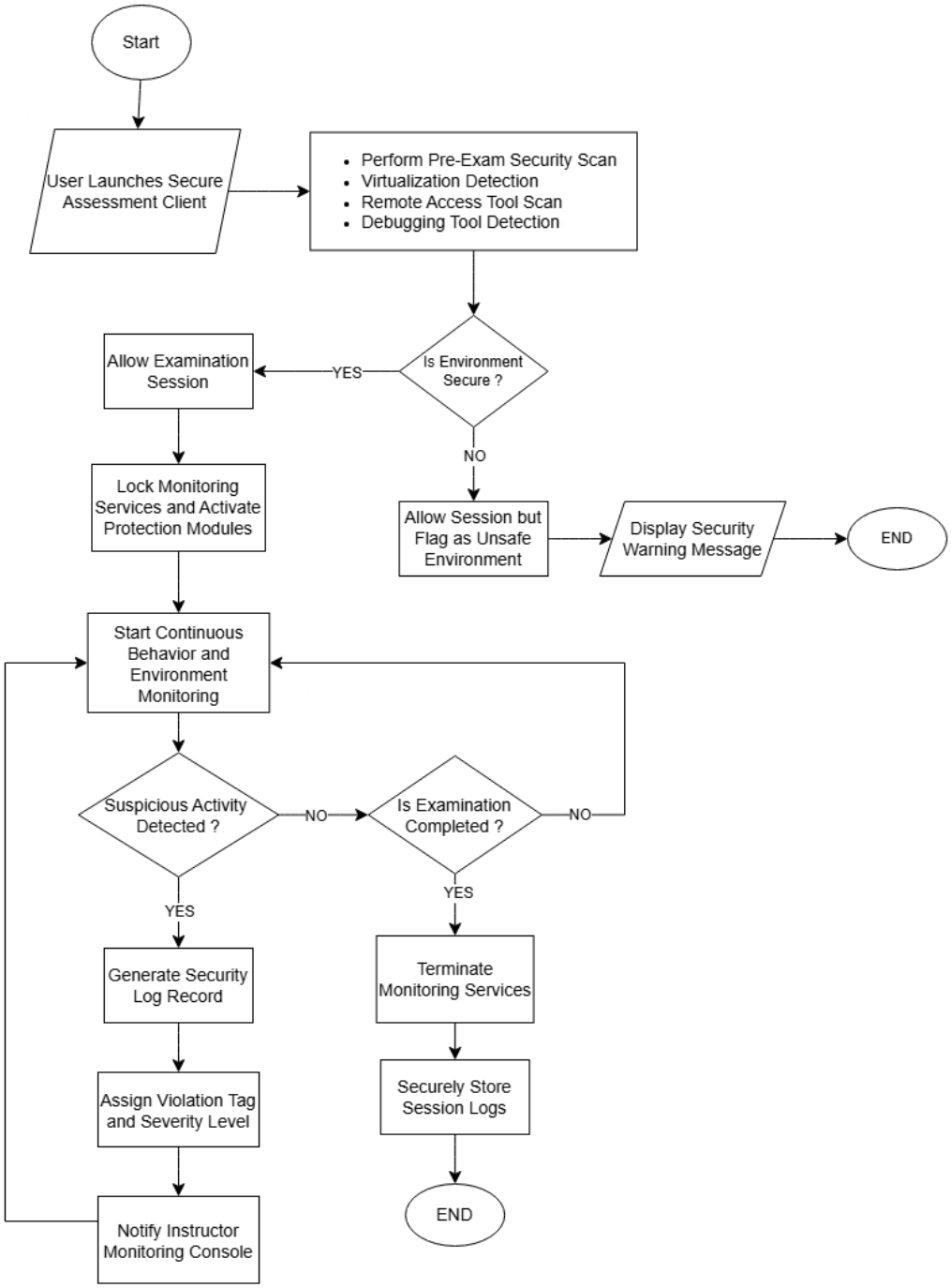


Figure B.10. **Detection Effectiveness Measurement Process**

APPENDIX C

Title